



RFL-RFT-C

RFT TRAINER KIT

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. RF & Microwave is one of the important subjects that need to be taught while learning Electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

⌚ Design ⌚ Quality ⌚ Value ⌚ Delivery ⌚ Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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INSTRUCTION

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to the packing list. Also ensure that no visible damage as such appears on any accessories.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions' sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions' bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS FOR RFL-RFT-C

INTRODUCTION:-

RF Circuit Training System introduce the concept of RF Transmission Line Circuits and Microwave Network Analysis. The RFT-C Board consists 13 modules of Transmission Channel & Microstrip Discontinuities. RF Trainer kit has been design at UHF freq bands. It is designed to experimentally analyze the characteristics of microstrip line for high speed transmission.

The trainer kit boosts the learning curve & visualization towards critical impedance theory along with microstrip transmission techniques and how some of their characteristics affect due to discontinuity in microstrip line. This system provides insight on RF Transmission line concepts through insertion loss based exercise. This system covers variety of RF measurement like insertion loss, return Loss, impedances, frequency response over wideband etc. This system is compatible with various measuring equipment like Spectrum Analyzer, Network Analyzer, RF Generator & Detector supplied as option with this kit Individual characterizations of device will help to understand S -parameters. The PCB is gold plated to provide better reliability & electrical performance and integrated board platform.

GENERAL SPECIFICATIONS:-

- Transmission Line : Microstrip Line
- Characteristics Impedance : 50 Ω
- Substrate material : FR4 with dielectric constant 4.4
- Substrate thickness : 1.6mm
- Copper thickness : 35 μ m
- Power handling capacity : +20dBm
- Connector : SMA Type
- Dimension : 350mm(W) $\times$ 330mm(L) $\times$ 70mm(H)
- Weight : 1.6 kg per board
- Mounting : Wooden box with acrylic

TECHNICAL SPECIFICATIONS:-

RFL-RFT-C: TRANSMISSION CHANNEL & MICROSTRIP DISCONTINUITIES

CPW with ground

- Frequency : 0.3GHz to 3GHz
- Power : 0dBm to -20dBm
- Microstrip line length : 46mm
- Track width : 3mm
- Insertion loss : < -3dB

Bend discontinuity

- Frequency : 0.3GHz to 1.3GHz
- Power : 0dBm to -20dB
- Microstrip line length : 46mm
- Track width : 3mm
- Insertion loss : < -3dB

Tee discontinuity

- Frequency : 0.3GHz to 1.3GHz
- Track width : 3mm
- Load Condition : Matched

Coupled microstrip line

- Frequency : 0.3GHz to 3GHz
- Power : 0dBm to -20dBm
- Microstrip line length : 58mm
- Track width : 3mm
- Coupling gap : 0.3mm
- Insertion loss : < -3dB

Via discontinuity

- Frequency : 0.3GHz to 1.3GHz
- Power : 0dBm to -20dB
- Microstrip line length : 43mm
- Track width : 3mm
- Via radius : 0.5mm
- Insertion loss : < -3dB

GAP discontinuity

- Frequency : 0.3GHz to 1.3GHz
- Power : 0dBm to -20dB
- Microstrip line length : 43mm
- Track width : 3mm
- Coupling gap : 0.25mm

Microstrip transmission line

- Frequency : 0.3GHz to 3GHz
- Power : 0dBm to 20dBm
- Microstrip line length : 43mm
- Track width : 3mm
- Insertion loss : < -3dB

Open discontinuity

- Frequency : 1.8GHz to 3GHz
- Power : 0dBm to -20dBm
- Microstrip line length : 43mm
- Track width : 3mm

Step discontinuity

- Frequency : 0.3GHz to 1.3GHz
- Power : 0dBm to -20dB
- Microstrip line length : 43mm
- Track width : 3mm

CPW No Ground transmission

- Frequency : 0.3GHz to 3GHz
- Power : 0dBm to -20dBm
- Microstrip line length : 52mm
- Track width : 2.4mm

Slot line

- Frequency : 0.3GHz to 1.3GHz
- Power : 0dBm to -20dBm
- Slot length : 62mm
- Slot width : 0.3mm

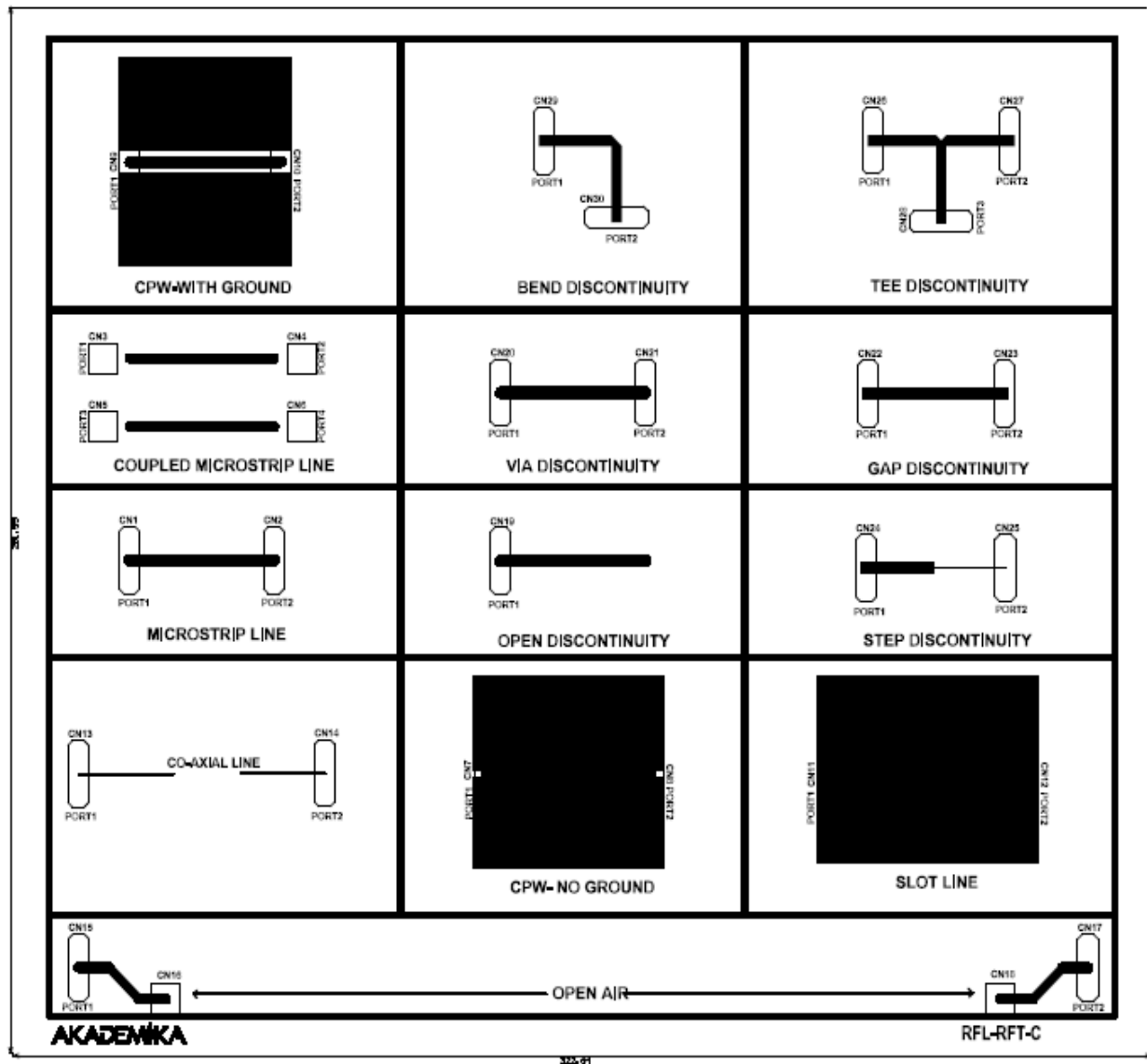
Co-axial cable transmission

- Frequency : 0.3GHz to 3GHz
- Power : 0dBm to -20dBm
- Coaxial cable length : 30cm
- Cable Type : RG 316
- Impedance : 50 ohm
- Insertion loss : < -3dB

Open air transmission

- Frequency : 0.8GHz to 1.1GHz
- Power : 0dBm to -20dBm
- Load : Matched Monopole Antenna
- Distance bet Tx and Rx : 24 cm

FUNCTIONAL BLOCK



EXPERIMENT NO.1

EXPERIMENT NO: 1

CPW-WITH GROUND Characterization

AIM:-

To understand the purpose and operational principle of the CPW-WITH GROUND by using the RFL-RFT-C module

OBJECTIVES:-

To explore the properties of CPW-WITH GROUND.

EQUIPMENTS:-

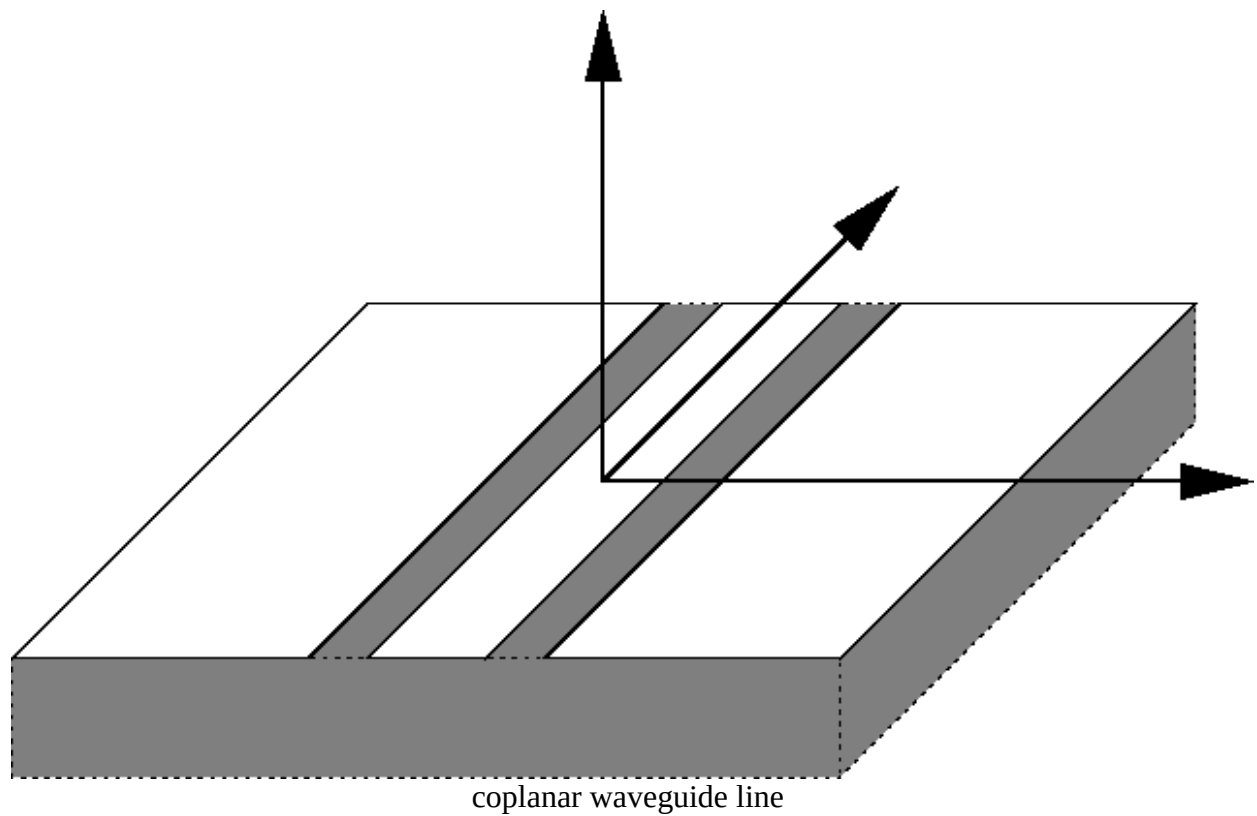
- RFL-RFT-C KIT
- Spectrum Analyzer

THEORY:-

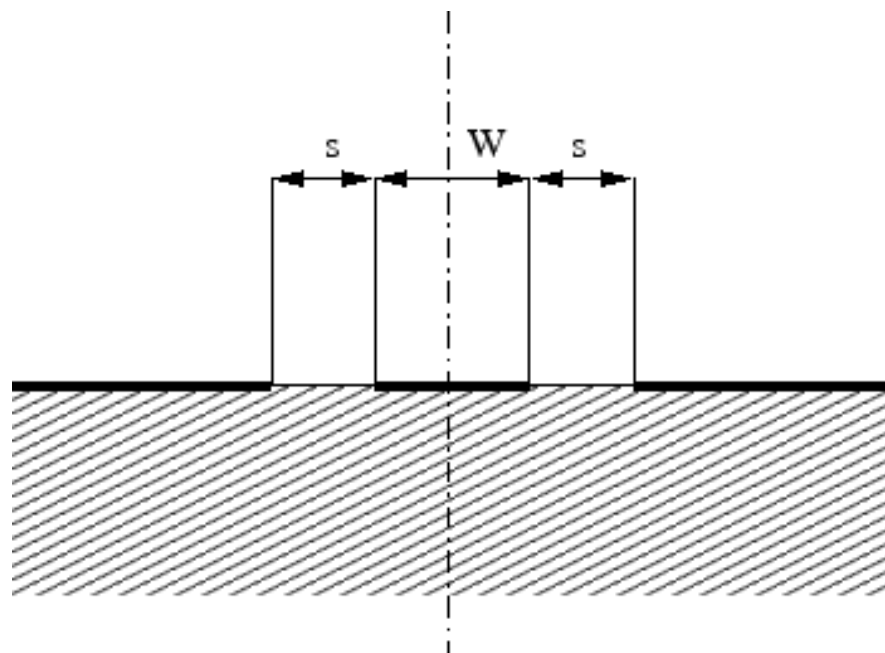
Coplanar Waveguide (CPW):

Definition

A coplanar line is a structure in which all the conductors supporting wave propagation are located on the same plane, i.e. generally the top of a dielectric substrate. There exist two main types of coplanar lines: the first, called coplanar waveguide (CPW), that we will study here, is composed of a median metallic strip separated by two narrow slits from an infinite ground plane, as may be seen on the figure below.

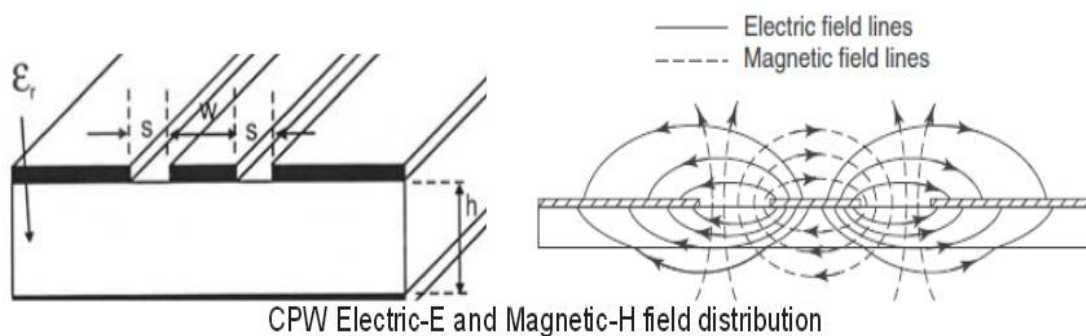


The characteristic dimensions of a CPW are the central strip width W and the width of the slots s . The structure is obviously symmetrical along a vertical plane running in the middle of the central strip.

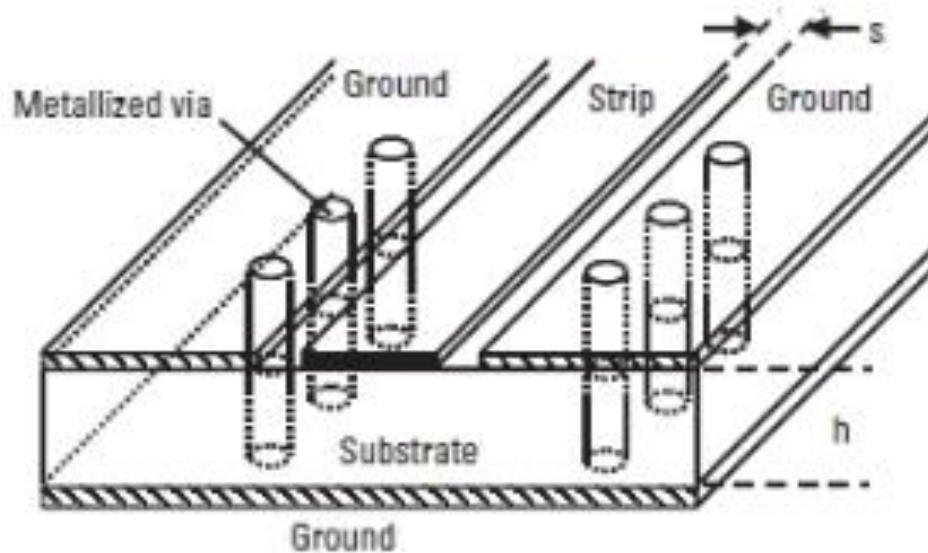


The other coplanar line, called a coplanar slot (CPS) is the complementary of that topology, consisting of two strips running side by side.

- The conductors formed a center strip separated by a narrow gap from two ground planes on either side. The dimensions of the center strip, the gap, the thickness and permittivity of the dielectric substrate determined the effective dielectric constant, characteristic impedance and the attenuation of the line.
- The gap in the coplanar waveguide is usually very small and supports electric fields primarily concentrated in the dielectric. With little fringing field in the air space, the coplanar waveguide exhibits low dispersion. In order to concentrate the fields in the substrate area and to minimize radiation, the dielectric substrate thickness is usually set equal to about twice the gap width.
- CPW has a zero cut-off frequency (suitable for wideband), but its low order propagation mode is indicated with Quasi-TEM because it is not a real TEM mode. At higher frequencies, the field becomes less-TEM, and more TE in nature. The CPW magnetic field is elliptically polarized.
- CPW it is printed circuit analogs of the three-wire transmission lines.



- Grounded Coplanar Waveguide (GCPW) is used on printed circuit boards as an alternative to Microstrip line. The gaps between the strip and ground is usually more than the thickness h of the substrate, so the GCPW field is concentrated between the strip and the substrate ground plane, and GCPW behaves like Microstrip. With vias connecting the ground planes, GCPW is less prone to radiate and has higher isolation than Microstrip.



- Since the number of the electric and magnetic field lines in the air is higher than the number of the same lines in the Microstrip case, the effective dielectric constant ϵ_{eff} of CPW is typically 15% lower than the ϵ_{eff} for Microstrip, so the maximum reachable characteristic impedance values are higher than the Microstrip values.
- The effect of finite dielectric substrate is almost ignorable if h exceeds $2b = W + 2s$.
- In addition, to avoid field radiation in the air, it is very important to use substrates with a high dielectric constant, with recommended values greater than 10, so that the electromagnetic field is mainly concentrated inside the dielectric.
- In CPW a ground plane exists between any two adjacent lines, hence cross talk effects between adjacent lines are very weak. As a result, CPW circuits can be made denser than conventional Microstrip circuits.

SET UP & PROCEDURE:-

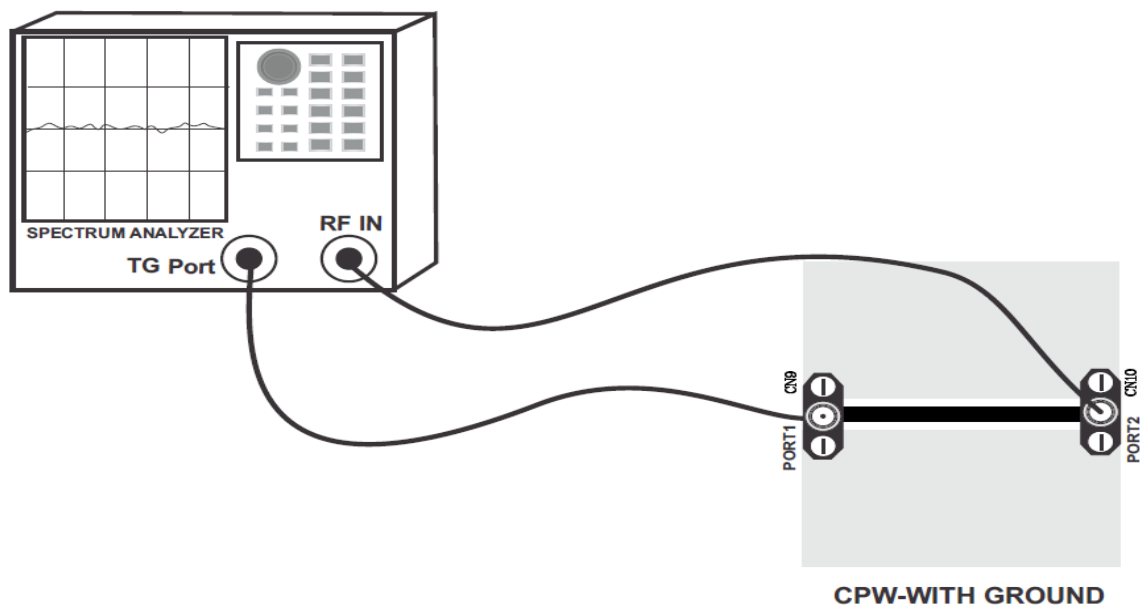
MEASUREMENT USING SPECTRUM ANALYZER:-

Measurement of Insertion Loss (S_{21} & S_{12}) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0dBm

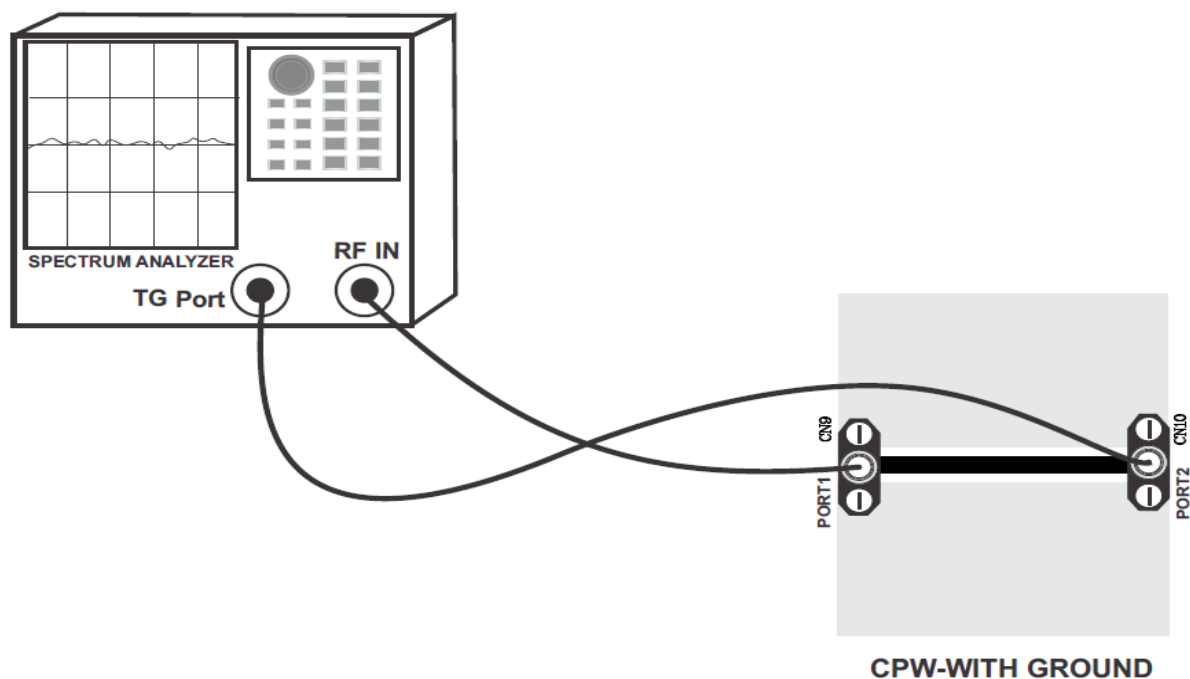
IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.*



Measurement Setup for insertion loss (S_{21})

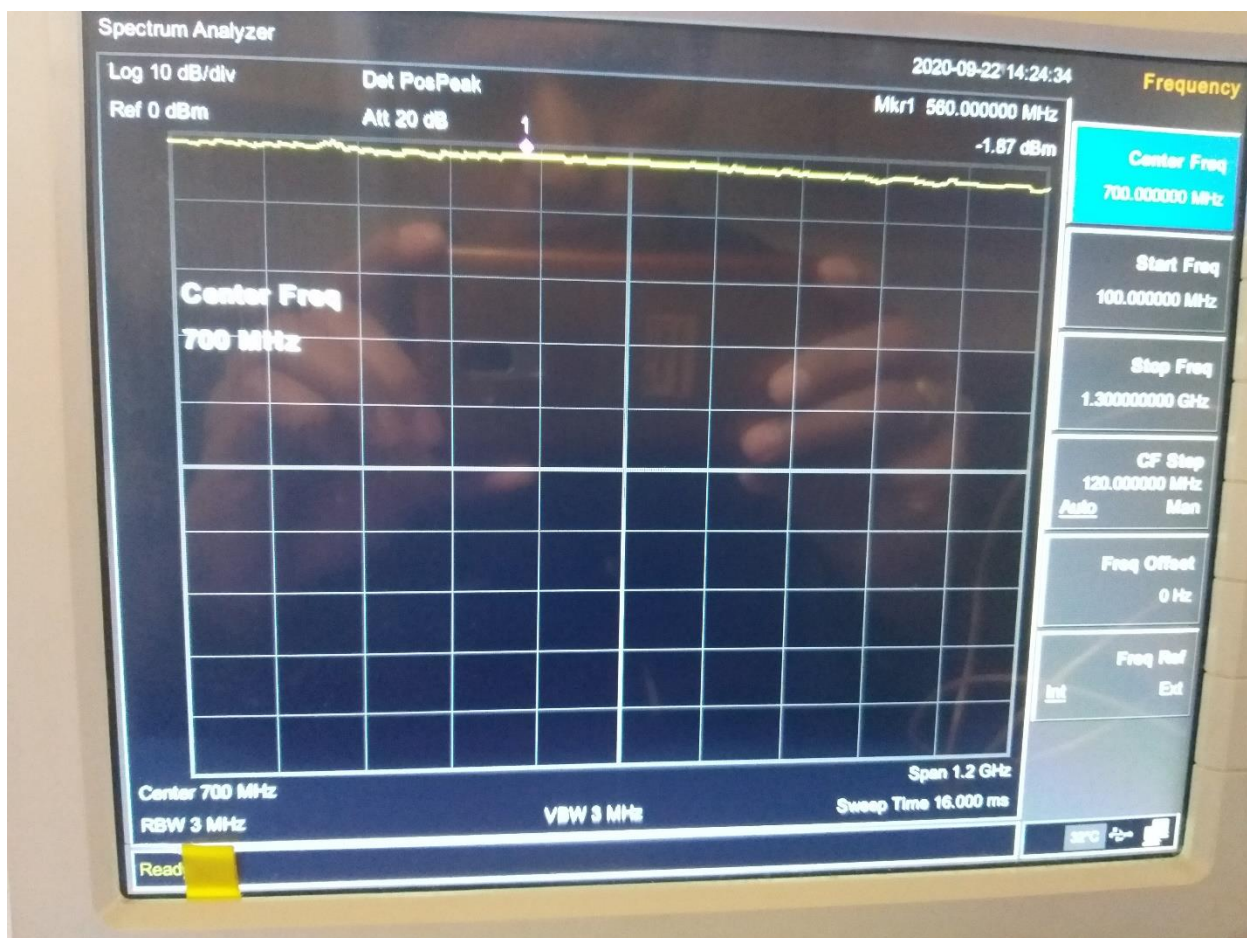
Measurement of Insertion Loss characteristics (S21):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyser to The CN9 (Port 1) of the CPW-WITH GROUND (i.e. signal from the spectrum Analyser is applied as input to the CPW-WITH GROUND)
- 2) Connect the CN10 (Port 2) of the CPW-WITH GROUND module to the Spectrum analyser Port 2 (i.e. output from the CPW-WITH GROUND is applied to the Spectrum analyser data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Measurement of Insertion loss characteristics (S12):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyser to The CN10 (Port 2) of the CPW-WITH GROUND (i.e. signal from the spectrum Analyser is applied as input to the CPW-WITH GROUND)
- 2) Connect the CN9 (Port 1) of the CPW-WITH GROUND module to the Spectrum analyser Port 2 (i.e. output from the CPW-WITH GROUND is applied to the Spectrum analyser data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



EXPERIMENT NO.2

EXPERIMENT NO: 2

Theory and Experimentation of BEND DISCONTINUITY

OBJECTIVE:-

To comprehend the purpose and operational principle of the BEND DISCONTINUITY by using the RFL-RFT-C module

AIM:-

To explore the properties of the BEND DISCONTINUITY module.

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

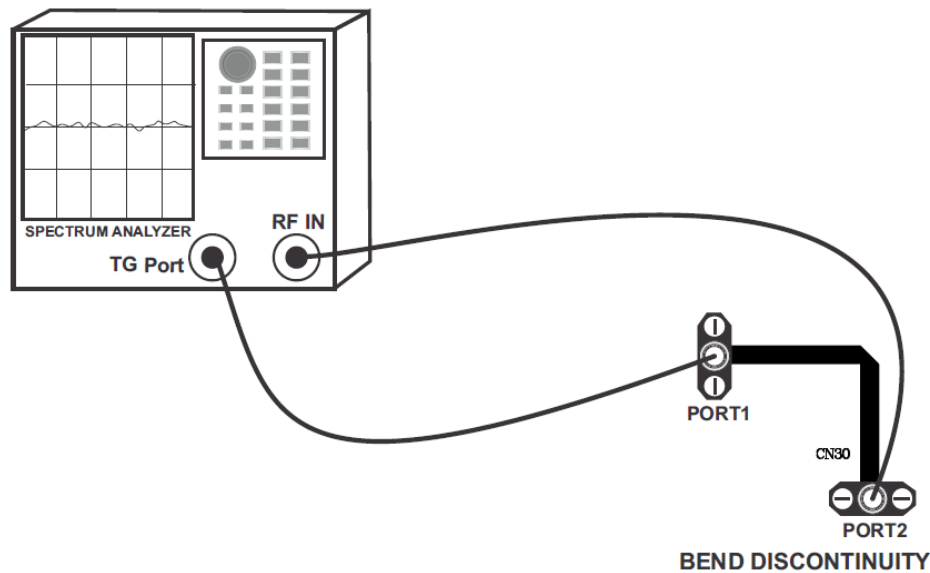
Using Spectrum Analyzer

Measurement of Insertion Loss (S21 & S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

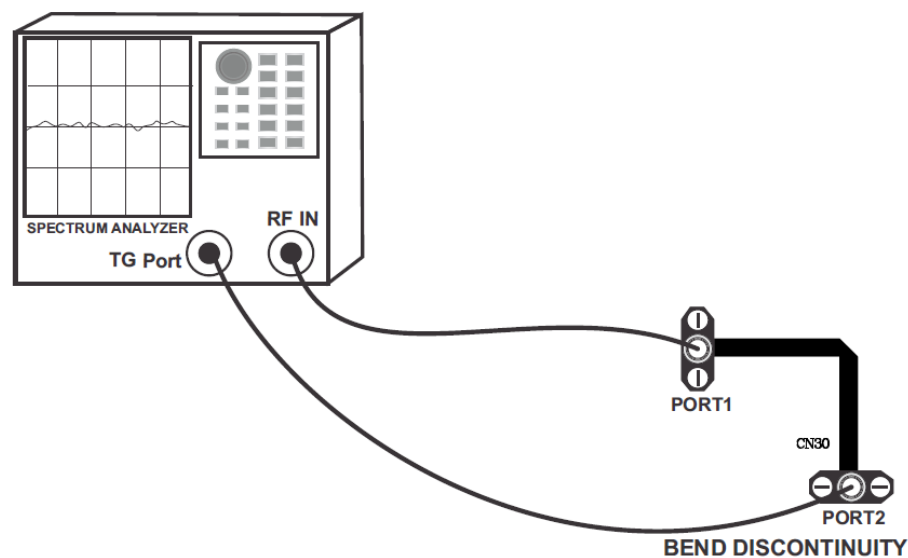
IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.*



Setup for insertion loss (S21)

Measurement of Insertion Loss characteristics (S21):

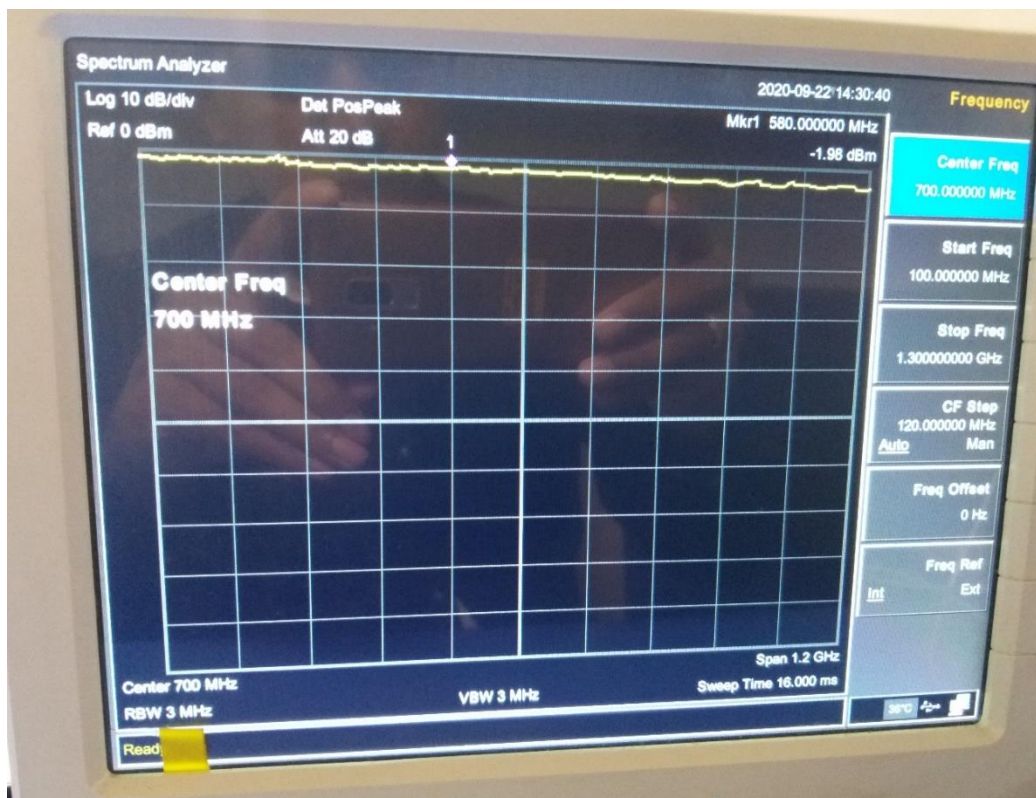
- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyser (Port1) to the CN29 of the BEND DISCONTINUTY module (i.e. signal of the Spectrum Analyzer is applied as input to the BEND DISCONTINUTY)
- 2) Connect the CN30 of the BEND DISCONTINUTY module to the Spectrum Analyzer Port 2 (i.e. output from the BEND DISCONTINUTY is applied to the Spectrum Analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.



Setup for insertion loss (S12)

- 1) Connect the output of the Spectrum analyzer (Port2) to the CN30 of the BEND DISCONTINUITY module (i.e. signal of the Spectrum Analyzer is applied as input to the BEND DISCONTINUITY)
- 2) Connect the CN29 of the BEND DISCONTINUITY module to the Spectrum Analyzer Port 2 (i.e. output from the BEND DISCONTINUITY is applied to the Spectrum Analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



**EXPERIMENT
NO.3**

EXPERIMENT NO: 3

Theory and Experimentation of TEE DISCONTINUITY

AIM:-

To comprehend the purpose and operational TEE DISCONTINUITY using the RFL-RFT-C module

OBJECTIVES:-

To explore the properties of TEE DISCONTINUITY and determination of insertion loss characteristics.

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

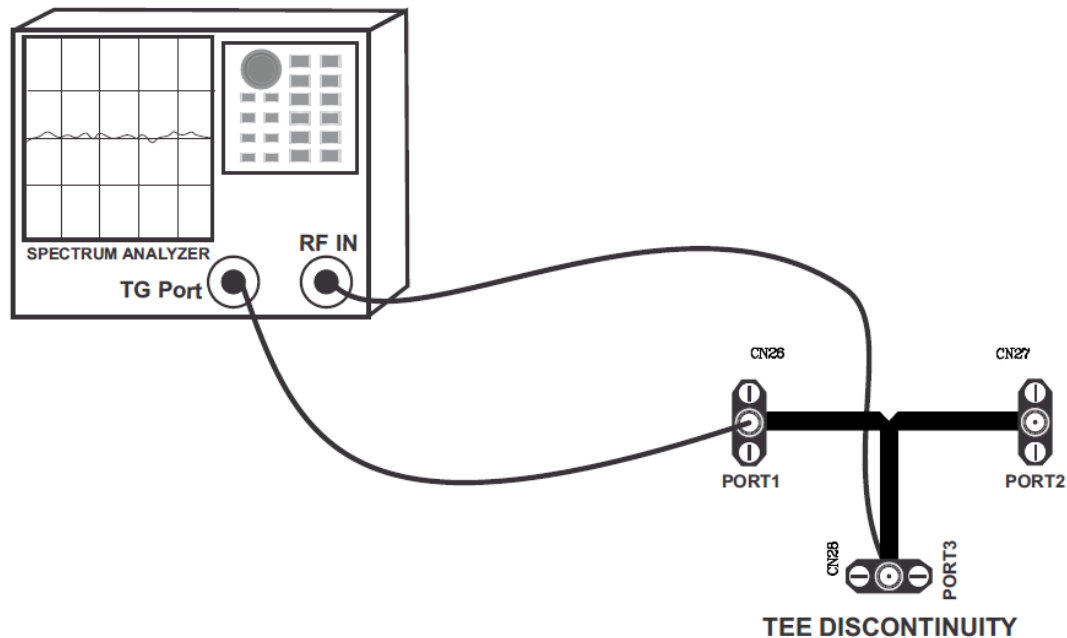
Using Spectrum Analyzer:-

Measurement of Insertion Loss (S21) Characteristics

Procedure:-

- Frequency Range ; 100 MHz-1.3 GHz
- Power Level : 0 dBm

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.*

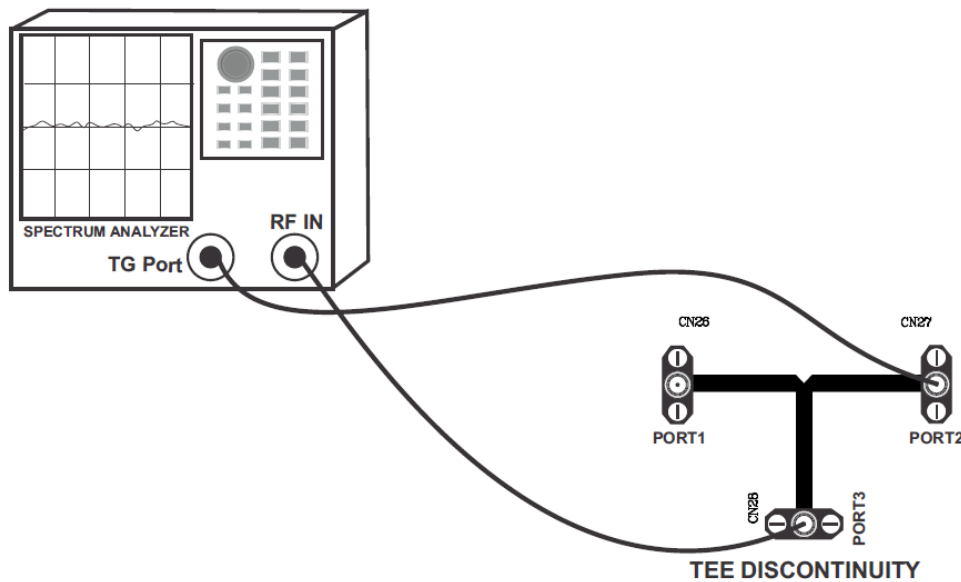


Setup for insertion loss (S31)

Measurement of Insertion Loss characteristics (S31):

Connection between Port 3 & Port 1;

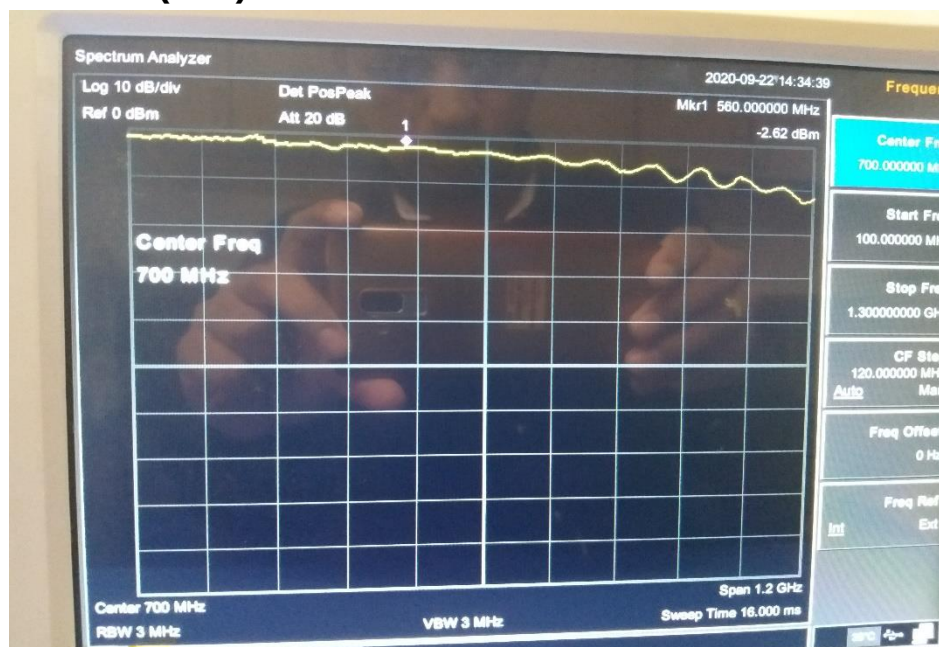
- 1) Connect the Tracking Generator (T.G. Port) output of the spectrum analyzer to the CN26 of the TEE DISCONTINUITY module (i.e. signal of the spectrum analyzer is applied to the TEE DISCONTINUITY)
- 2) Connect the CN28 of the TEE DISCONTINUITY module to the spectrum analyzer Port 2 (i.e. output from the TEE DISCONTINUITY is applied to the spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

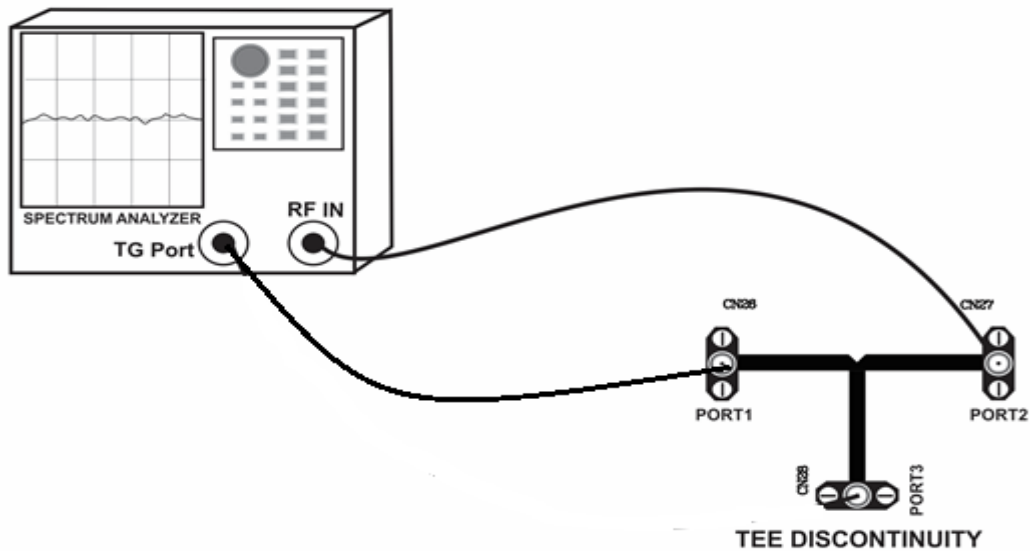


Setup for insertion loss (S32)

Connection between Port 3 & Port 2;

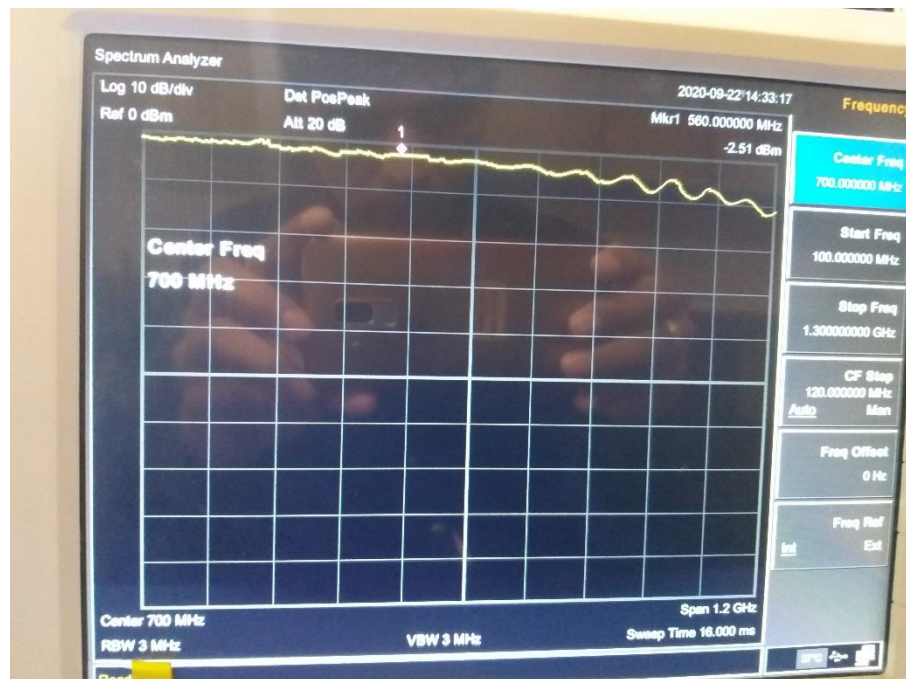
- 1) Connect the Tracking Generator (T.G. Port) output of the spectrum analyzer to the CN27 of the TEE DISCONTINUITY module (i.e. signal of the spectrum analyzer is applied as input to the TEE DISCONTINUITY)
- 2) Connect the CN28 of the TEE DISCONTINUITY module to the spectrum analyzer Port 2 (i.e. output from the TEE DISCONTINUITY is applied to the spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S31):

Measurement of Insertion loss characteristics (S21):**Connection between Port 1 & Port 2;**

- 1) Connect the Port 1 (CN26) of the TEE DISCONTINUITY module to the spectrum analyzer TG port (i.e. signal of the spectrum is applied as input to the TEE DISCONTINUITY).
- 2) Connect the output of the spectrum analyzer RF IN to the Port2 CN27 of the TEE DISCONTINUITY module (i.e. output from the TEE DISCONTINUITY is applied to the spectrum analyzer for data display)
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



EXPERIMENT NO.4

EXPERIMENT NO: 4

Theory and Experimentation of Coupled Microstrip Line.

OBJECTIVE:-

To comprehend the purpose and operational principle of the COUPLED MICROSTRIP LINE by using the RFL-RFT-C module

AIM:-

To explore the properties of the COUPLED MICROSTRIP LINE.

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

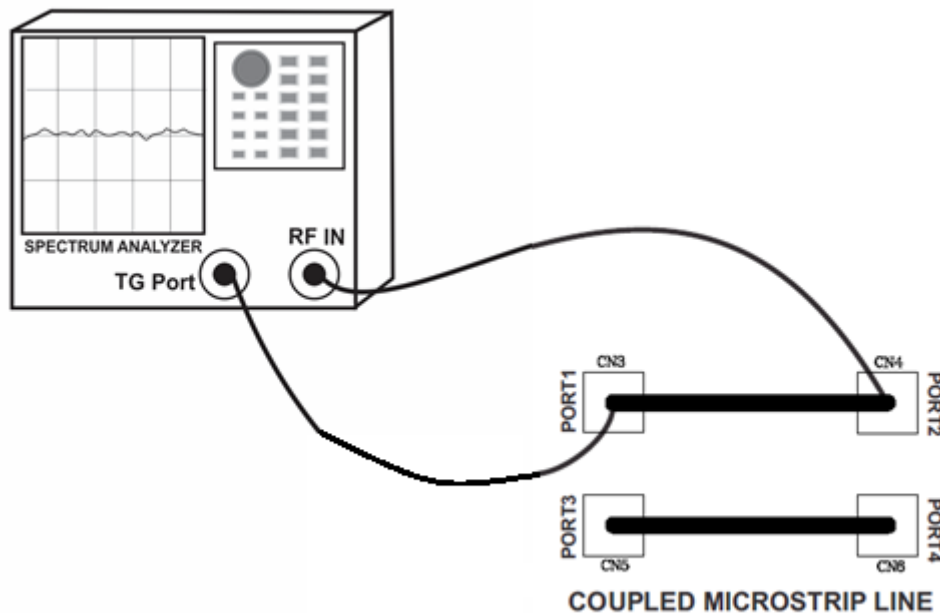
A. Using Spectrum Analyzer:-

Measurement of Insertion Loss (S21 &S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.



Setup for insertion loss (S21)

Measurement of Insertion Loss characteristics (S21):

Connection between Port 1 & Port 2;

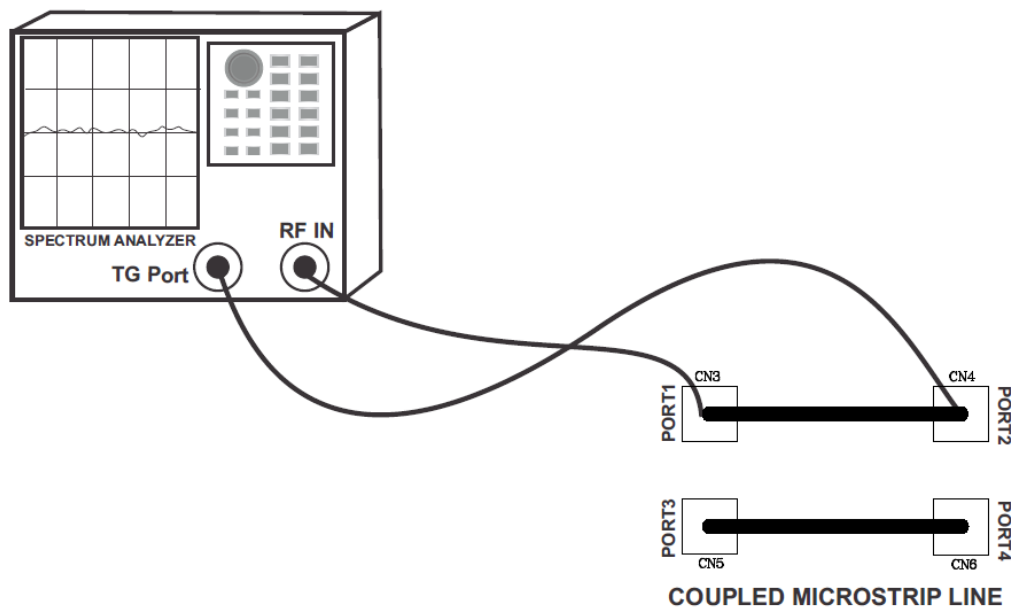
- 1) Connect the Tracking Generator (T.G. Port) output of the spectrum analyzer to the CN3 of the COUPLED MICROSTRIP LINE module (i.e. signal of the spectrum analyzer is applied as input to the COUPLED MICROSTRIP LINE)
- 2) Connect the CN4 of the COUPLED MICROSTRIP LINE module to the spectrum analyzer Port 2 (i.e. output from the COUPLED MICROSTRIP LINE applied to the VNA for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Observe the output of S12.

Connection between Port 1 & Port 3:

- 1) Connect the Tracking Generator (T.G. Port) output of the spectrum analyzer to the CN3 of the COUPLED MICROSTRIP LINE module (i.e. signal of the spectrum analyzer is applied as input to the COUPLED MICROSTRIP LINE)
- 2) Connect the CN5 of the COUPLED MICROSTRIP LINE module to the spectrum analyzer Port 2 (i.e. output from the COUPLED MICROSTRIP LINE applied to the spectrum analyzer for data display).
- 3) Connect 50 Ω Load to Port 2.
- 4) Apply markers and note down the readings for the frequencies specified in the observation table.
- 5) Save trace data for post processing and save trace image.

Connection between Port 1 & Port 4:

- 1) Connect the Tracking Generator (T.G. Port) output of the spectrum analyzer to the CN3 of the COUPLED MICROSTRIP LINE module (i.e. signal of the spectrum analyzer is applied as input to the COUPLED MICROSTRIP LINE)
- 2) Connect the CN6 of the COUPLED MICROSTRIP LINE module to the spectrum analyzer Port 2 (i.e. output from the COUPLED MICROSTRIP LINE applied to the spectrum analyzer for data display).
- 3) Connect 50 Ω Load to Port 2 & Port 3.
- 4) Apply markers and note down the readings for the frequencies specified in the observation table.
- 5) Save trace data for post processing and save trace image.

Measurement of Insertion loss characteristics (S12):**Connection between Port 1 & Port 2;**

- 1) Connect the Tracking Generator (T.G. Port) output of the spectrum analyzer to the CN4 of the COUPLED MICROSTRIP LINE module (i.e. signal of the spectrum analyzer is applied as input to the COUPLED MICROSTRIP LINE)
- 2) Connect the CN3 of the COUPLED MICROSTRIP LINE module to the spectrum analyzer Port 2 (i.e. output from the COUPLED MICROSTRIP LINE applied to the spectrum for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

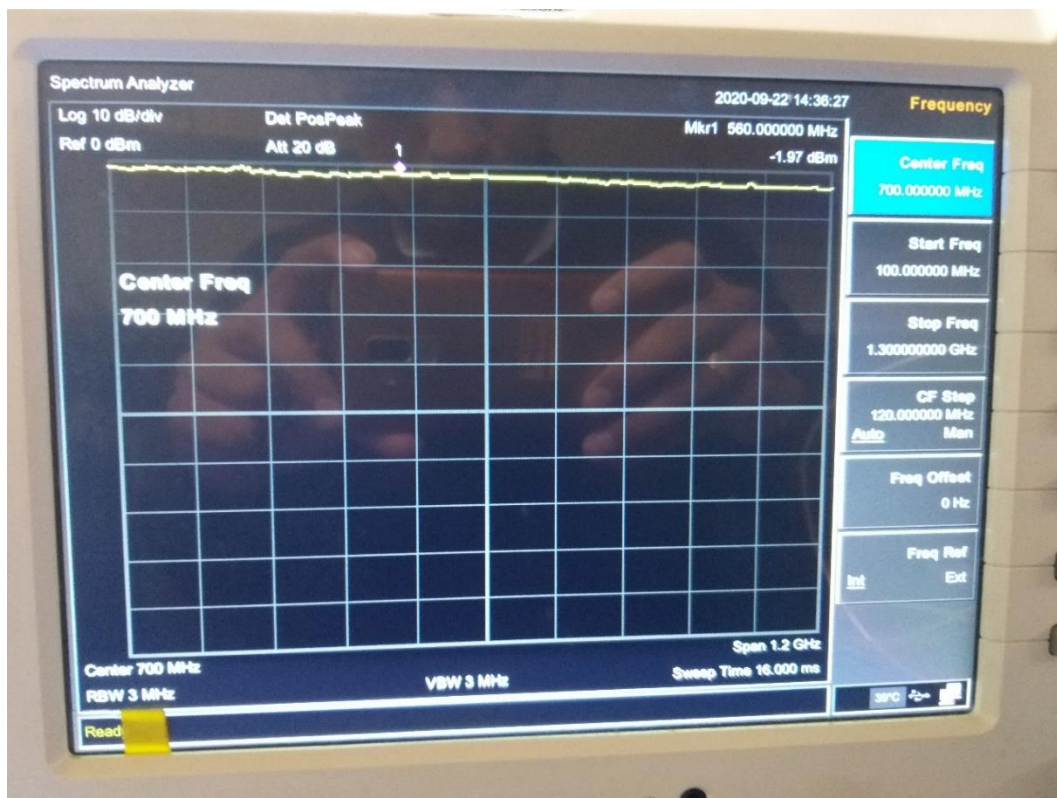
Connection between Port 1 & Port 3:

- 1) Connect the Tracking Generator (T.G. Port) output of the spectrum analyzer to the CN3 of the COUPLED MICROSTRIP LINE module (i.e. signal of the spectrum analyzer is applied as input to the COUPLED MICROSTRIP LINE)
- 2) Connect the CN5 of the COUPLED MICROSTRIP LINE module to the spectrum analyzer Port 2 (i.e. output from the COUPLED MICROSTRIP LINE applied to the spectrum analyzer for data display).
- 3) Connect 50 Ω Load to Port 2.
- 4) Apply markers and note down the readings for the frequencies specified in the observation table.
- 5) Save trace data for post processing and save trace image.

Connection between Port 1 & Port 4:

- 1) Connect the output of the spectrum analyzer (Port1) to the CN6 of the COUPLED MICROSTRIP LINE module (i.e. signal of the spectrum analyzer is applied as input to the COUPLED MICROSTRIP LINE)
- 2) Connect the CN3 of the COUPLED MICROSTRIP LINE module to the spectrum analyzer Port 2 (i.e. output from the COUPLED MICROSTRIP LINE applied to the spectrum analyzer for data display).
- 3) Connect 50 Ω Load to Port 2 & Port 3.
- 4) Apply markers and note down the readings for the frequencies specified in the observation table.
- 5) Save trace data for post processing and save trace image.

Reference Plot (S21):



EXPERIMENT No.5

EXPERIMENT NO.5

Theory and Experimentation of VIA DISCONTINUITY.

AIM:-

To comprehend the purpose and operational principle of the Via Discontinuity by using the RFL-RFT-C module

OBJECTIVES:-

To explore the properties of the Via Discontinuity.

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

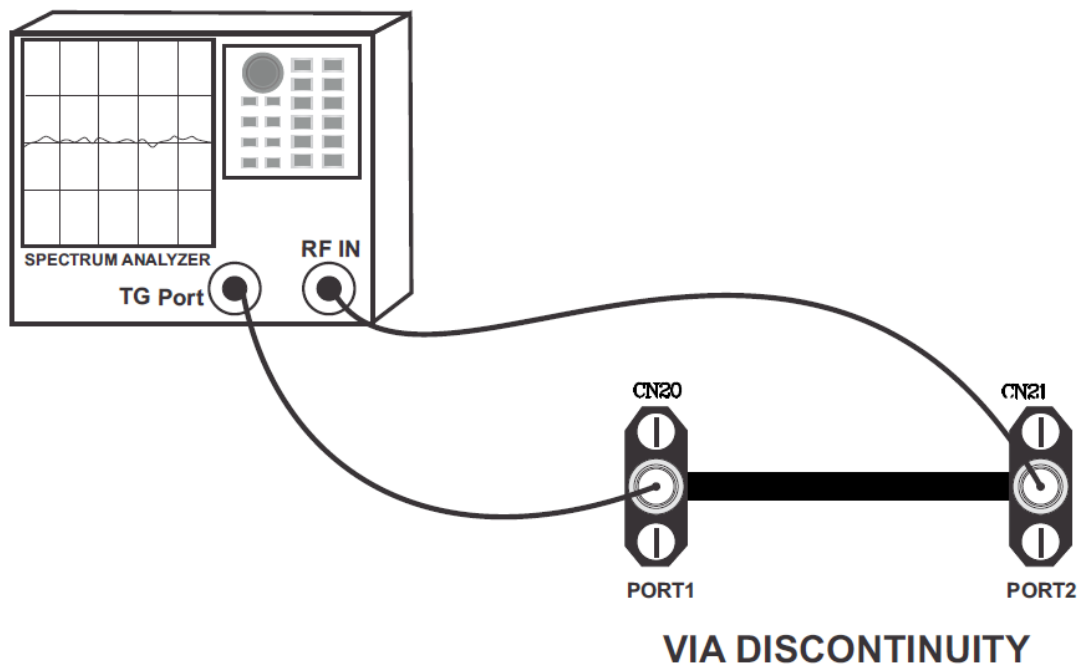
A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21 & S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.*



Setup for insertion loss (S21)

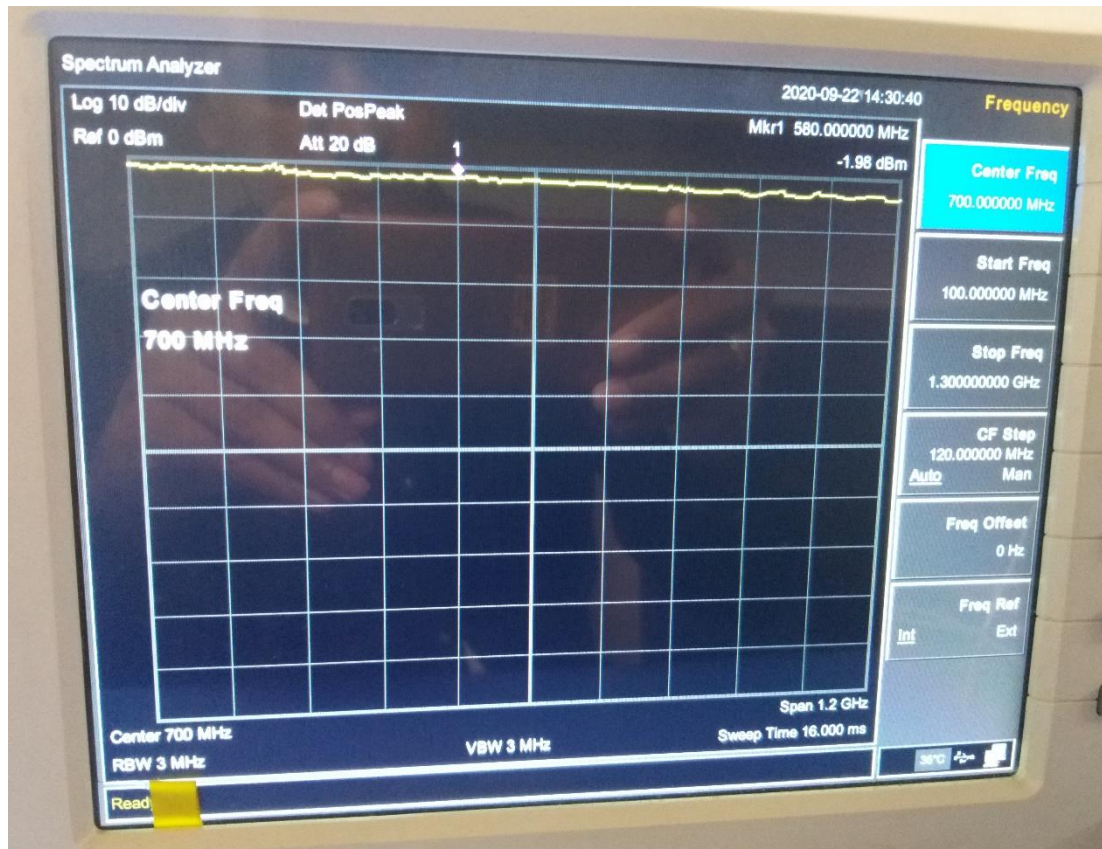
Measurement of Insertion Loss characteristics (S21):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the CN20 of the Via Discontinuity module (i.e. signal of the Spectrum analyzer is applied as input to the Via Discontinuity)
- 2) Connect the CN21 of the Via Discontinuity module to the Spectrum analyzer Port 2 (i.e. output from the Via Discontinuity is applied to the Spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Measurement of Insertion loss characteristics (S12):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the CN21 of the Via Discontinuity module (i.e. signal of the Spectrum analyzer is applied as input to the Via Discontinuity)
- 2) Connect the CN20 of the Via Discontinuity module to the Spectrum analyzer Port 2 (i.e. output from the Via Discontinuity is applied to the Spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



**EXPERIMENT
NO.6**

EXPERIMENT NO.6

Theory and Experimentation of GAP DISCONTINUITY

AIM:-

To understand the purpose and operational principle of the Gap Discontinuity by using the RFL-RFT-C module

OBJECTIVES:-

To explore the properties of the Gap Discontinuity.

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

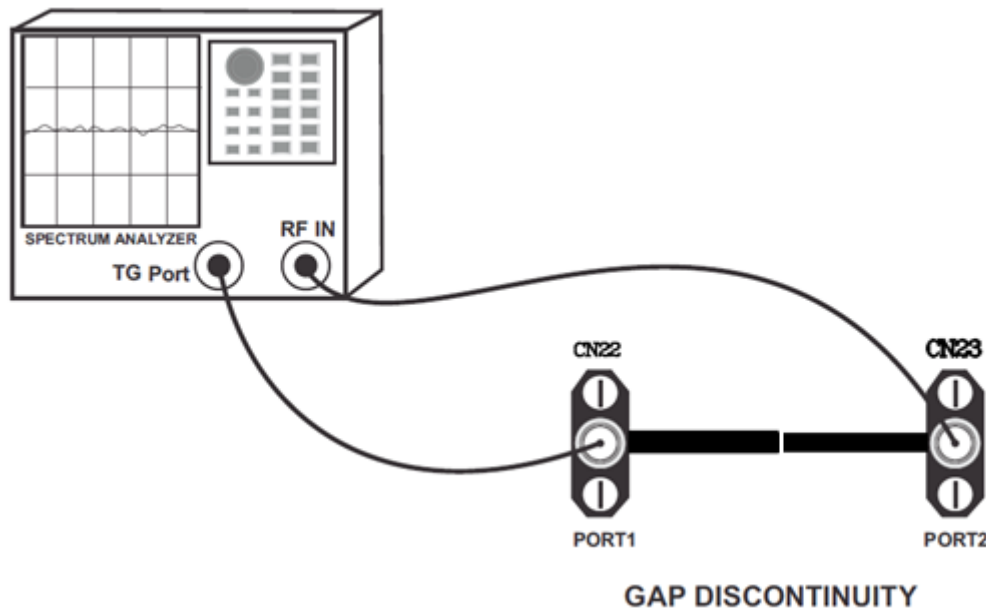
A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21 & S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.



Setup for insertion loss (S21)

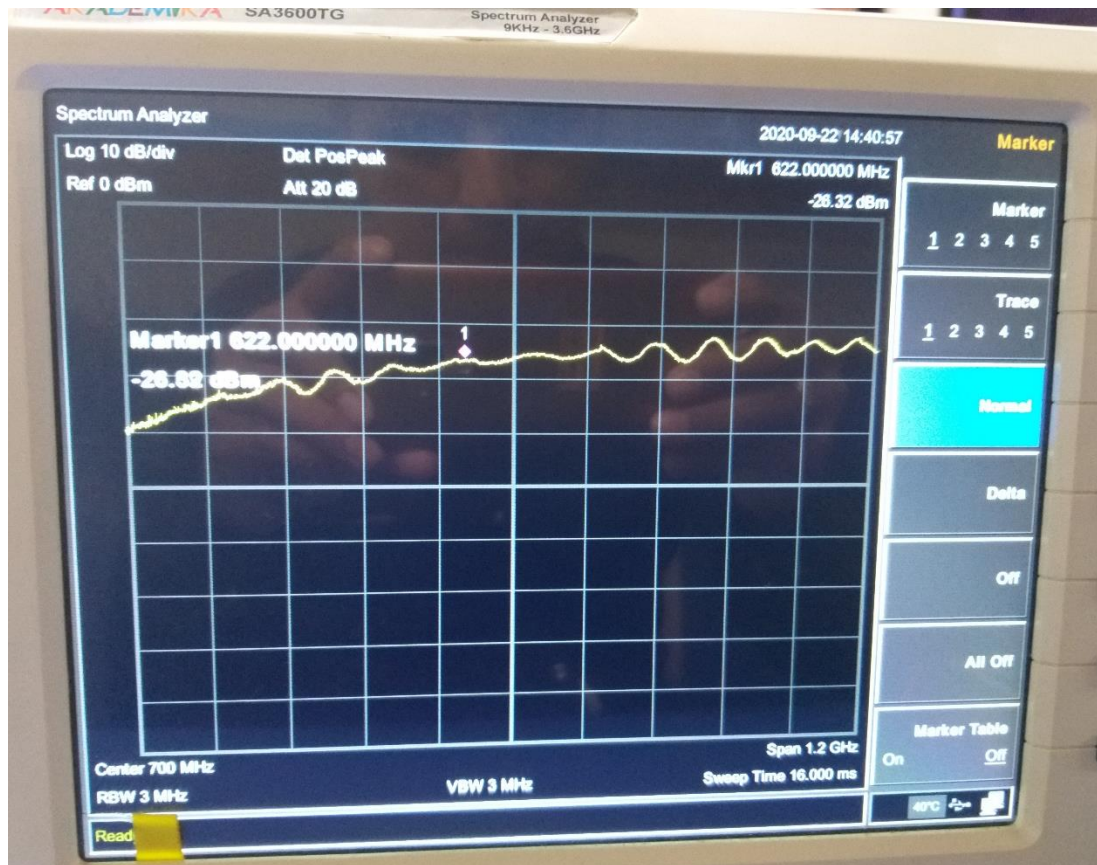
Measurement of Insertion Loss characteristics (S21):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the CN22 of the Gap Discontinuity module (i.e. signal of the Spectrum analyzer is applied as input to Gap Discontinuity)
- 2) Connect the CN23 of the Gap Discontinuity module to the Spectrum analyzer Port 2 (i.e. output from the Gap Discontinuity is applied to the Spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Measurement of Insertion loss characteristics (S12):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the CN23 of the Gap Discontinuity module (i.e. signal of the Spectrum analyzer is applied as input to Gap Discontinuity)
- 2) Connect the CN22 of the Gap Discontinuity module to the Spectrum Analyzer Port 2 (i.e. output from the Gap Discontinuity is applied to the Spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



**EXPERIMENT
NO.7**

EXPERIMENT NO.7

MICROSRTIP LINE CHARACTERIZATION

AIM:-

To understand the purpose and operational principle of the Microsrtip Line by using the RFL-RFT-C module

OBJECTIVES:-

To explore the properties of Microstrip Line.

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

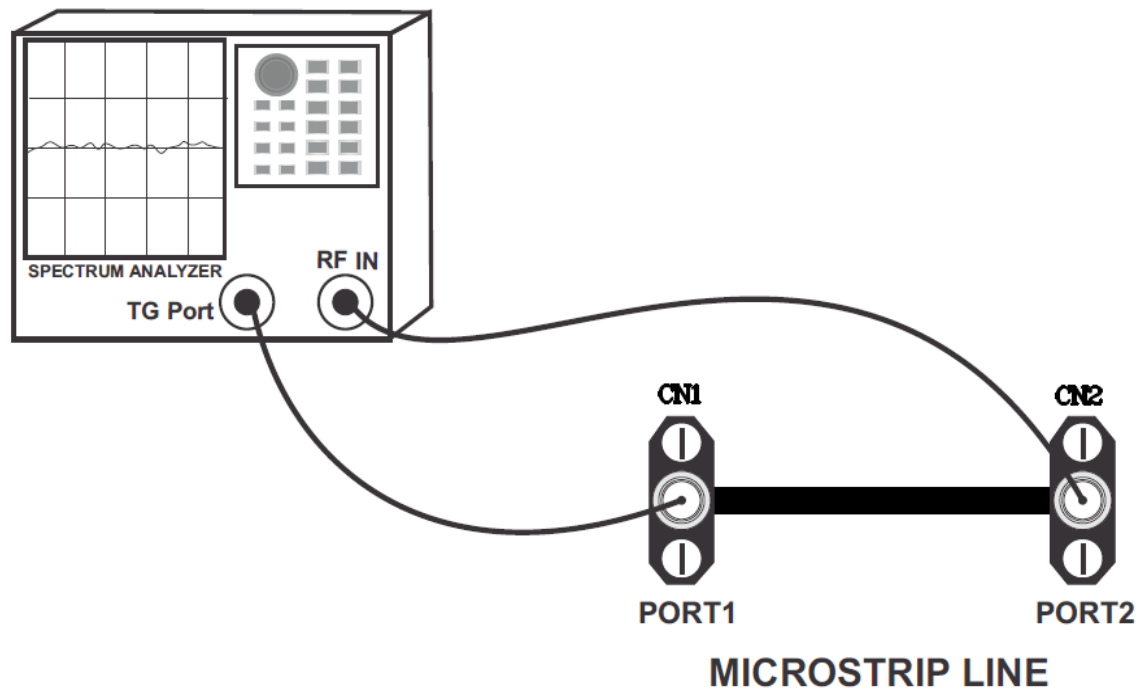
A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21 & S12) Characteristics

Procedure:-

- Frequency Range ; 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.*



Setup for insertion loss (S21)

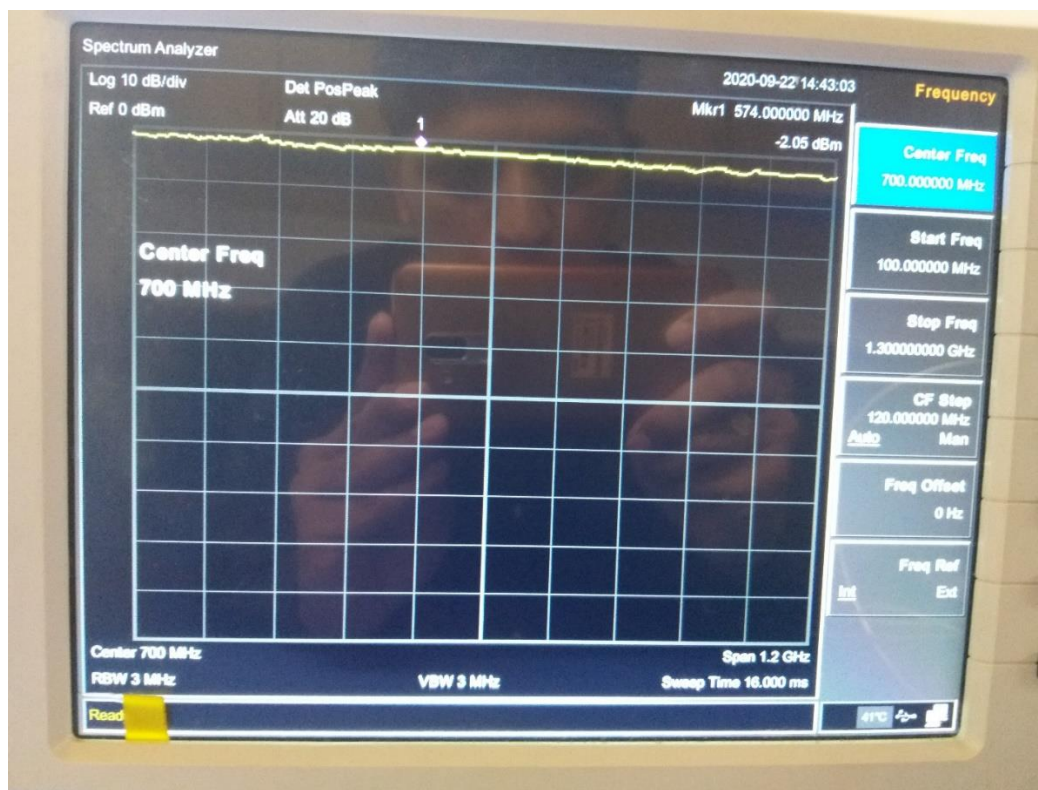
Measurement of Insertion Loss characteristics (S21):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN1(Port 1) of the Microstrip Line module (i.e. signal of the Spectrum analyzer is applied as input to the Microstrip Line)
- 2) Connect the CN2 of the Microstrip Line module to the Spectrum Analyzer Port 2 (i.e. output from the Microstrip Line is applied to the Spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Measurement of Through Port Insertion loss characteristics (S12):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN2 of the Microstrip Line module (i.e. signal of the Spectrum analyzer is applied as input to the Microstrip Line)
- 2) Connect the CN1 of the Microstrip Line module to the Spectrum analyzer Port 2 (i.e. output from the Microstrip Line is applied to the Spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



EXPERIMENT NO.8

EXPERIMENT NO. 8

OPEN DISCONTINUITY CHARACTERIZATION

AIM:-

To understand the purpose and operational principle of the Open Discontinuity by using the RFL-RFT-C module

OBJECTIVE:-

To explore the properties of Open Discontinuity.

EQUIPMENTS:-

- RFL-RFT-C KIT
- VNA

SET UP & PROCEDURE:-

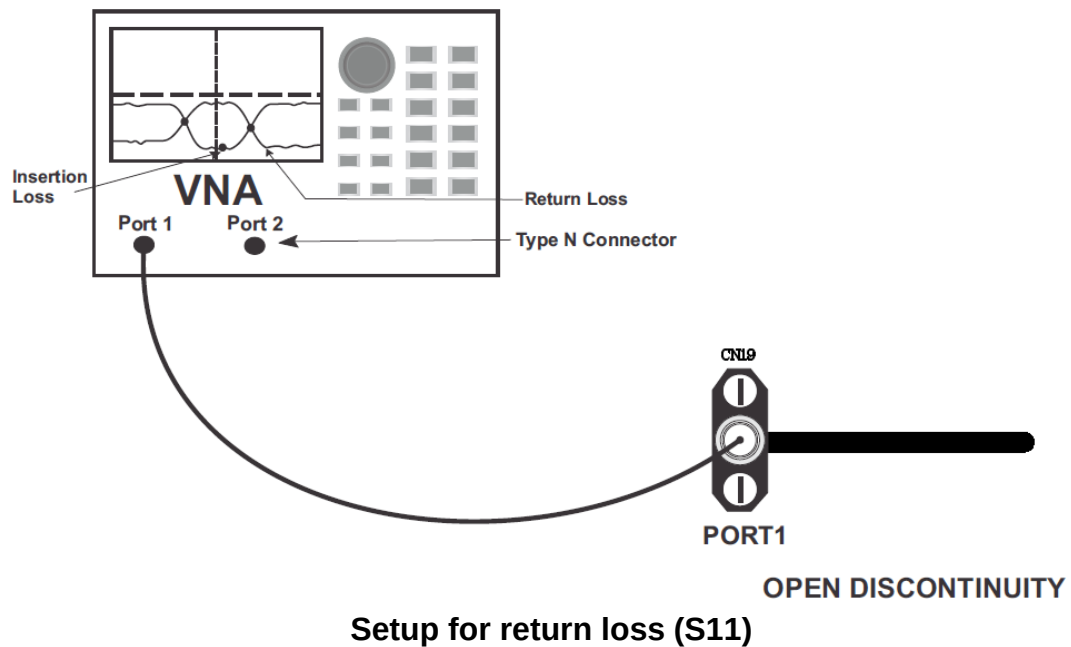
A. Procedure Using VNA

Measurement of return loss (S11) Characteristics

Procedure:-

- Frequency Range : 300 MHz to 3.0 GHz
- Power Level : 0 dBm or lower (eg. -5dBm, -10 dBm)

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.



Measurement of Return Loss characteristics (S11):

- 1) Connect the output of the network analyzer (Port1) to the CN19 of the Open Discontinuity module (i.e. signal of the VNA is applied as input to the Open Discontinuity)
- 2) Apply markers and note down the readings for the frequencies specified in the observation table.
- 3) Save trace data for post processing and save trace image.

EXPERIMENT NO.9

EXPERIMENT NO. 9

STEP DISCONTINUITY CHARACTERIZATION

AIM:-

To understand the purpose and operational principle of the STEP DISCONTINUITY by using the RFL-RFT-C module

OBJECTIVES:-

To explore the properties of Step Discontinuity

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

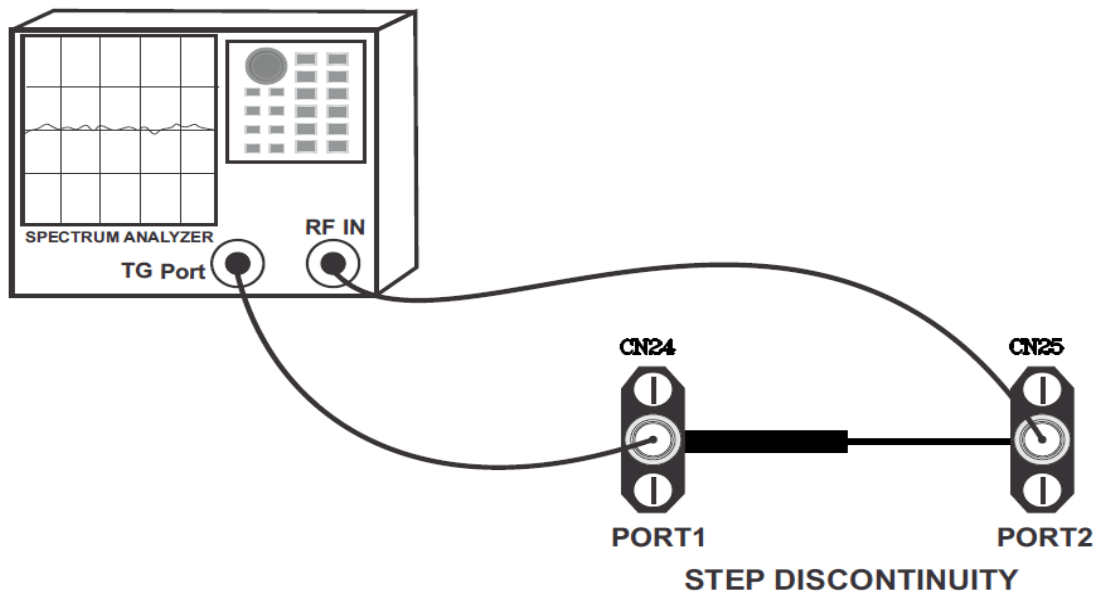
A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21 and S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.*



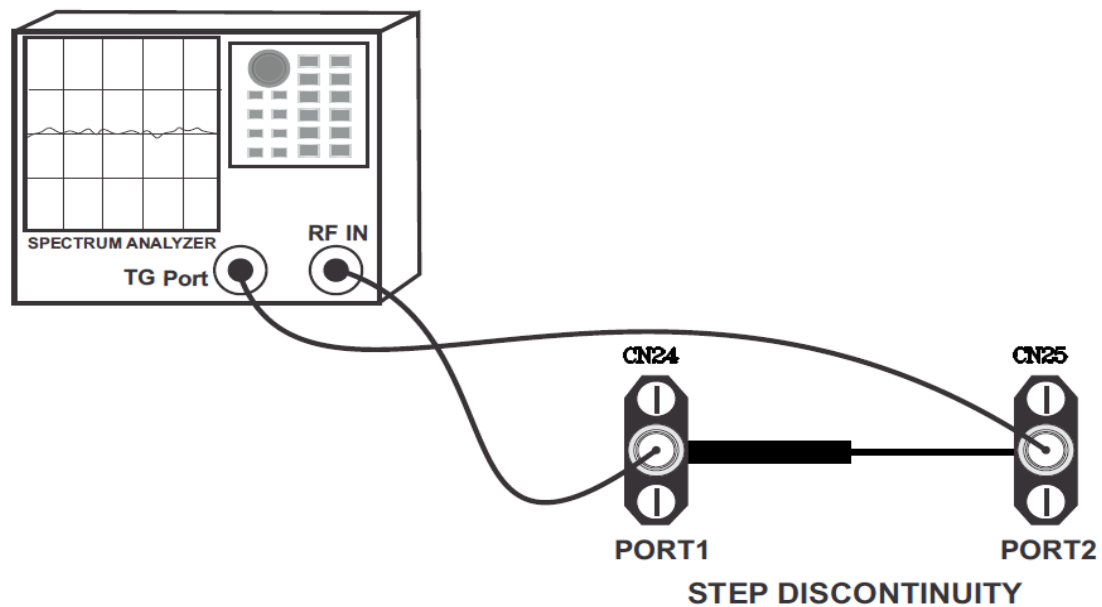
Setup for insertion loss (S21)

Measurement of Insertion Loss characteristics (S21):

- 1) Connect the TG port of the Spectrum Analyzer (Port1) to the CN24(Port1) of the Step Discontinuity module (i.e. signal of the Spectrum Analyzer is applied as input to the Step Discontinuity)
- 2) Connect the CN25 (Step 2) of the Step Discontinuity module to the Spectrum Analyzer Port 2 (i.e. output from the Step Discontinuity is applied to the Spectrum Analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

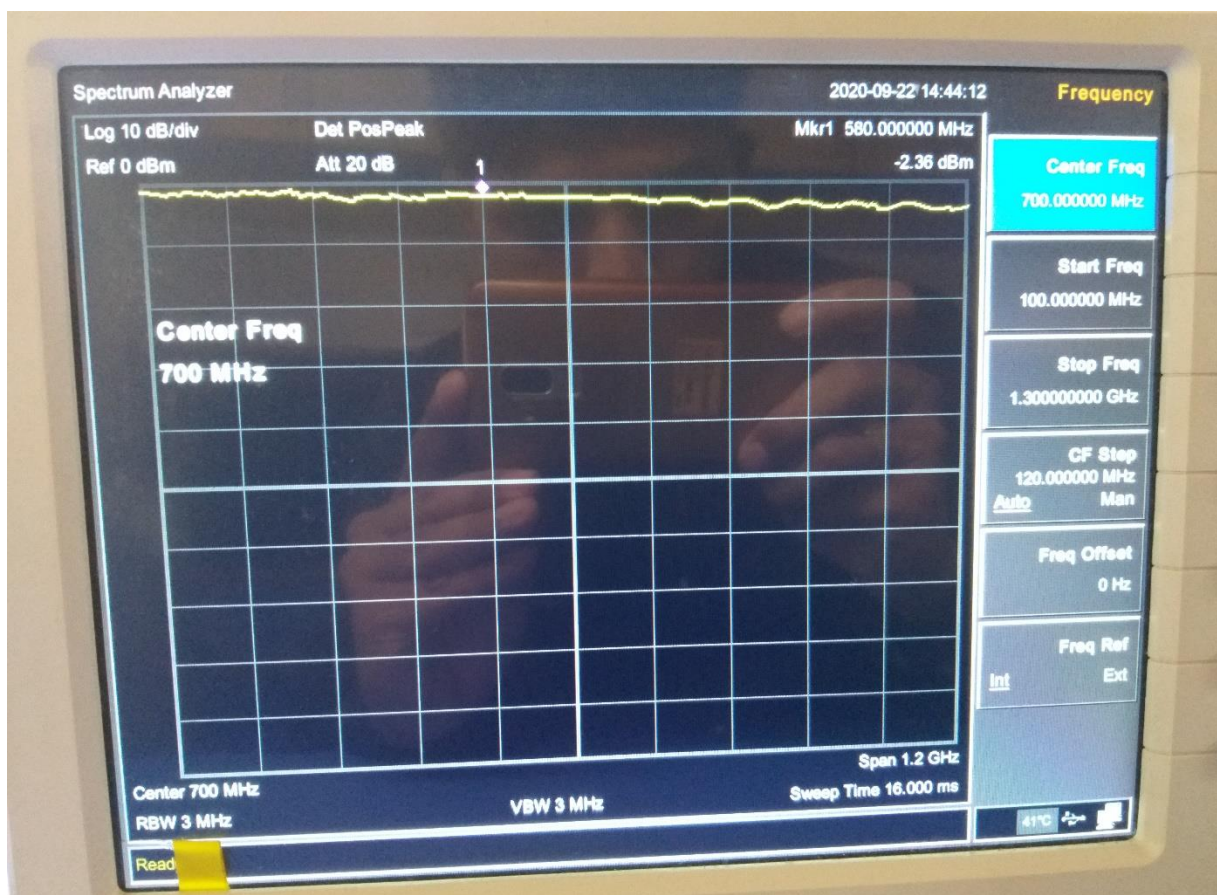
Measurement of Insertion loss characteristics (S12):

- 1) Connect the output of the Spectrum Analyzer (Port1) to the CN25(Port2) of the Step Discontinuity module (i.e. signal of the Spectrum Analyzer is applied as input to the Step Discontinuity)
- 2) Connect the CN24 (Step 1) of the Step Discontinuity module to the Spectrum Analyzer Port 2 (i.e. output from the Step Discontinuity is applied to the Spectrum Analyzer for data display).



- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



**EXPERIMENT
NO.10**

EXPERIMENT NO. 10

CO-AXIAL LINE CHARACTERIZATION

AIM:-

To understand the purpose and operational principle of the CO-Axial Line by using the RFL-RFT-C module

OBJECTIVES:-

To explore the properties of Co-Axial Line.

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

THEORY:-

SET UP & PROCEDURE:-

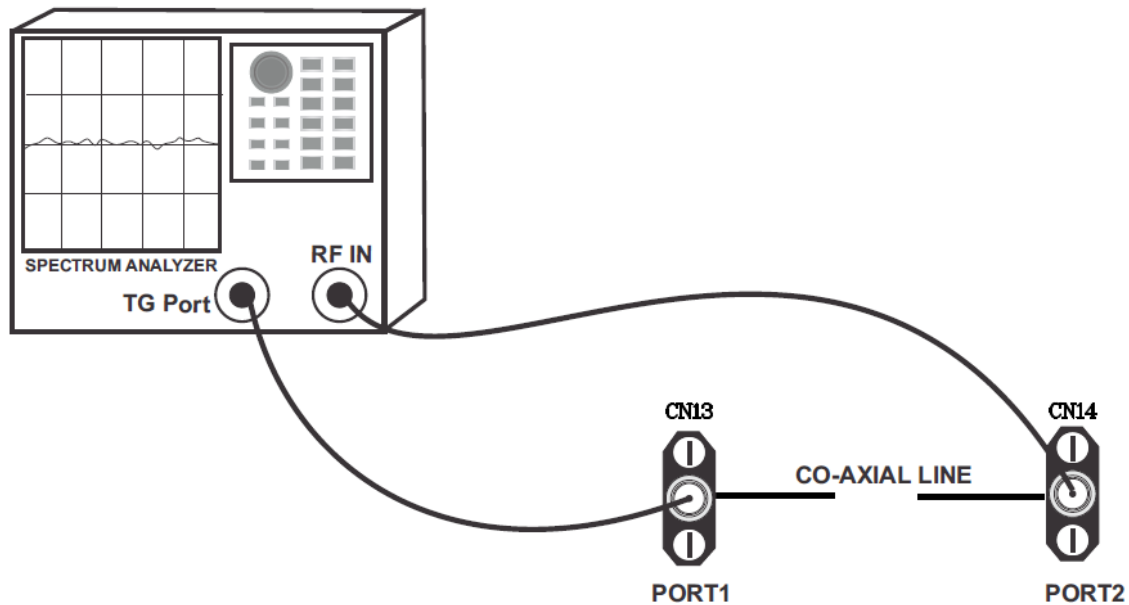
A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21 & S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.*



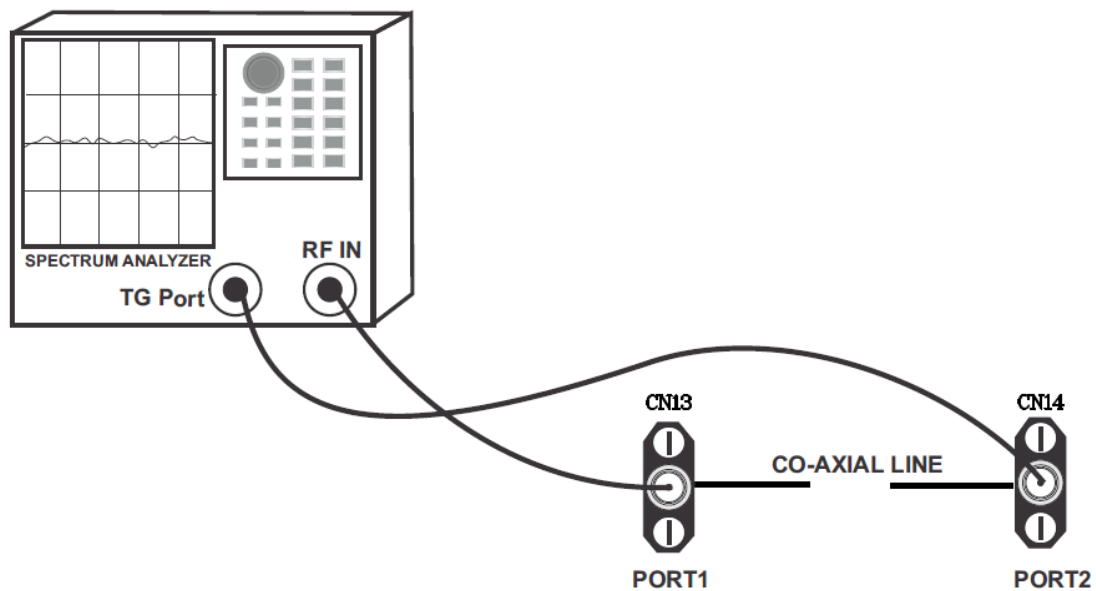
Setup for insertion loss (S21)

Measurement of Insertion Loss characteristics (S21):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN13(Port 1) of the Co-Axial Line module (i.e. signal of the Spectrum Analyzer is applied as input to the Co-Axial Line)
- 2) Connect the CN14 (Port 2) of the Co-Axial Line module to the Spectrum Analyzer Port 2 (i.e. output from the Co-Axial Line is applied to the Spectrum Analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

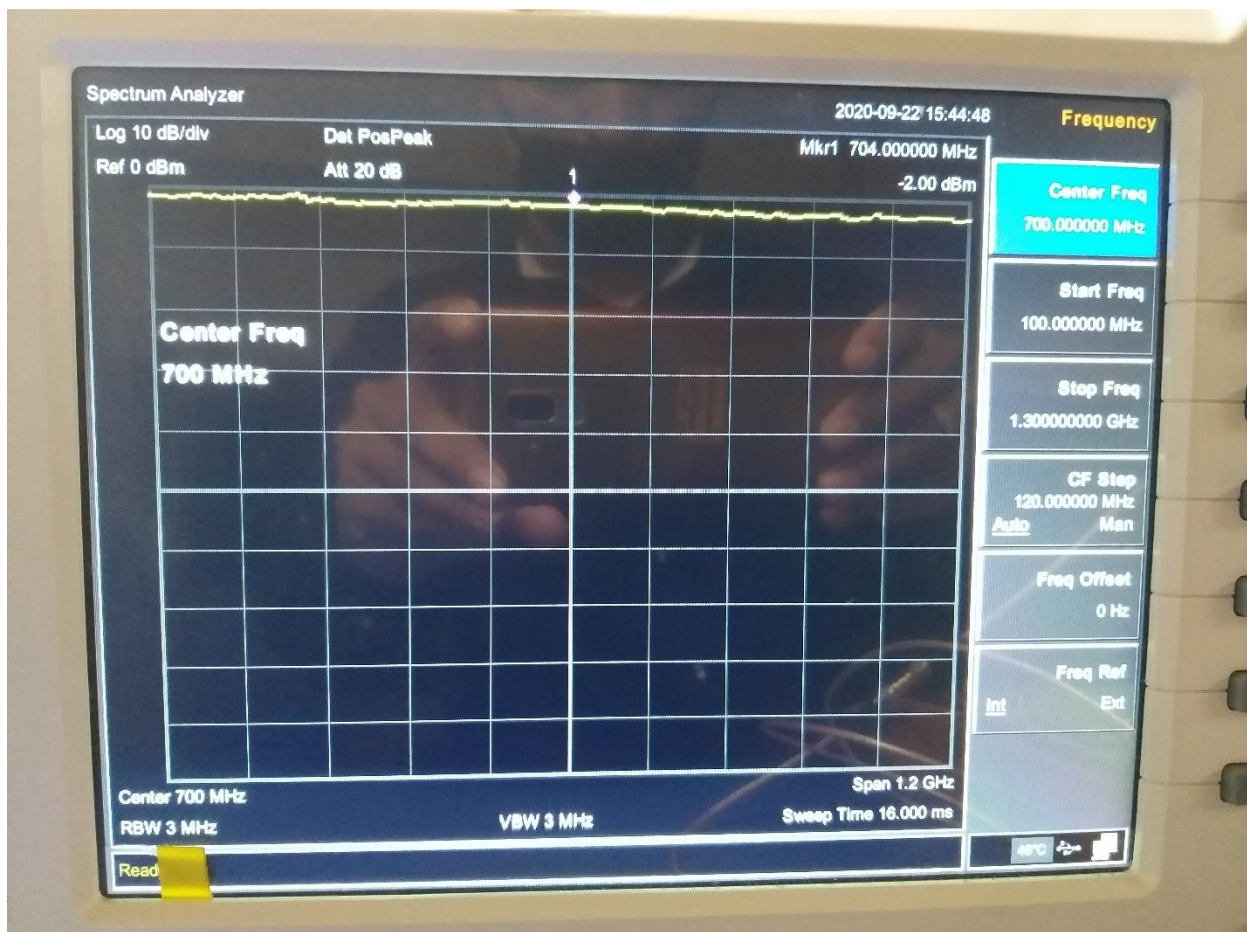
Measurement of Insertion loss characteristics (S12):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN14(Port 2) of the Co-Axial Line module (i.e. signal of the Spectrum Analyzer is applied as input to the Co-Axial Line)
- 2) Connect the CN13(Port 1) of the Co-Axial Line module to the Spectrum analyzer Port 2 (i.e. output from the Co-Axial Line is applied to the Spectrum Analyzer for data display).



- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S₂₁):



**EXPERIMENT
NO.11**

EXPERIMENT NO. 11

CHARACTERISATION OF CPW-NO GROUND

AIM:-

To understand the purpose and operational principle of the CPW-NO GROUND circuits by using the RFL-RFT-C modules

OBJECTIVES:-

To explore the properties of CPW-No Ground

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

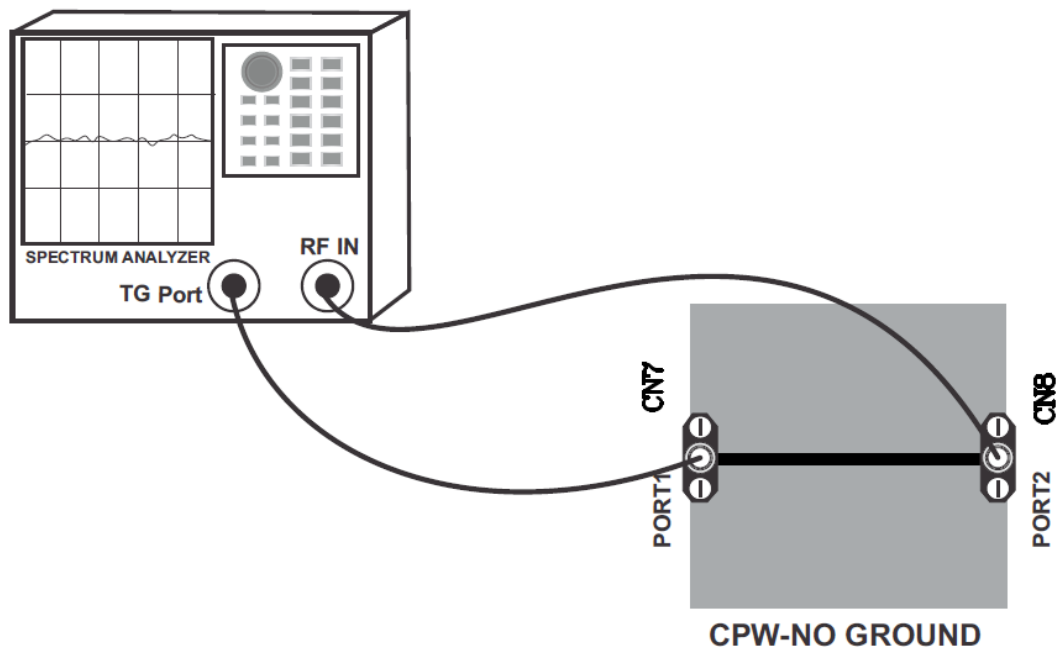
A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21 & S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.



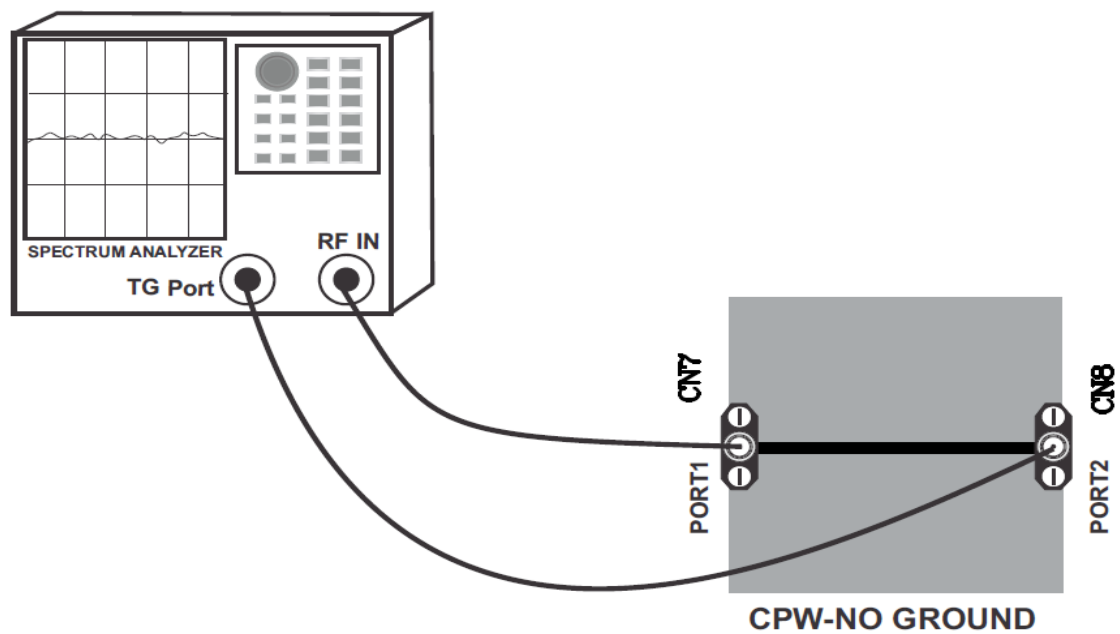
Setup for insertion loss (S21)

Measurement of Insertion Loss characteristics (S21):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN7 of the CPW-NO GROUND module (i.e. signal of the Spectrum analyzer is applied as input to the CPW-NO GROUND)
- 2) Connect the CN8 of the CPW-NO GROUND module to the Spectrum analyzer Port 2 (i.e. output from the CPW-NO GROUND is applied to the Spectrum analyzer for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

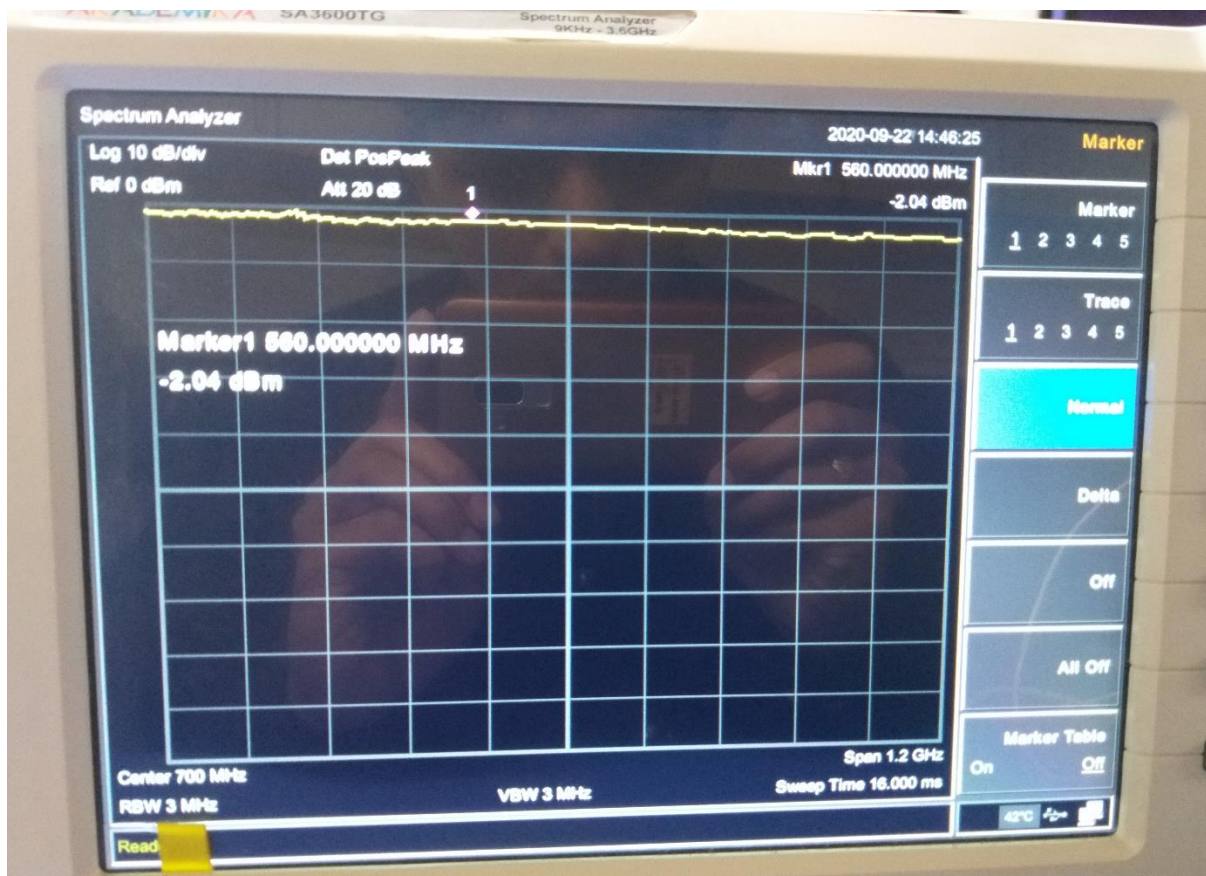
Measurement of Insertion loss characteristics (S12):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN8 of the CPW-NO GROUND module (i.e. signal of the Spectrum analyzer is applied as input to the CPW-NO GROUND)
- 2) Connect the CN7 of the CPW-NO GROUND module to the Spectrum analyzer Port 2 (i.e. output from the CPW-NO GROUND is applied to the Spectrum analyzer for data display).



- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reference Plot (S21):



**EXPERIMENT
NO.12**

EXPERIMENT NO. 12

CHARACTERISATION OF SLOT LINE

AIM:-

To understand the purpose and operational principle of the SLOT LINE circuits by using the RFL-RFT-C modules

OBJECTIVES:-

To explore the properties of Slot Line,

EQUIPMENTS:-

- RFL-RFT-C KIT
- Spectrum Analyzer

SET UP & PROCEDURE:-

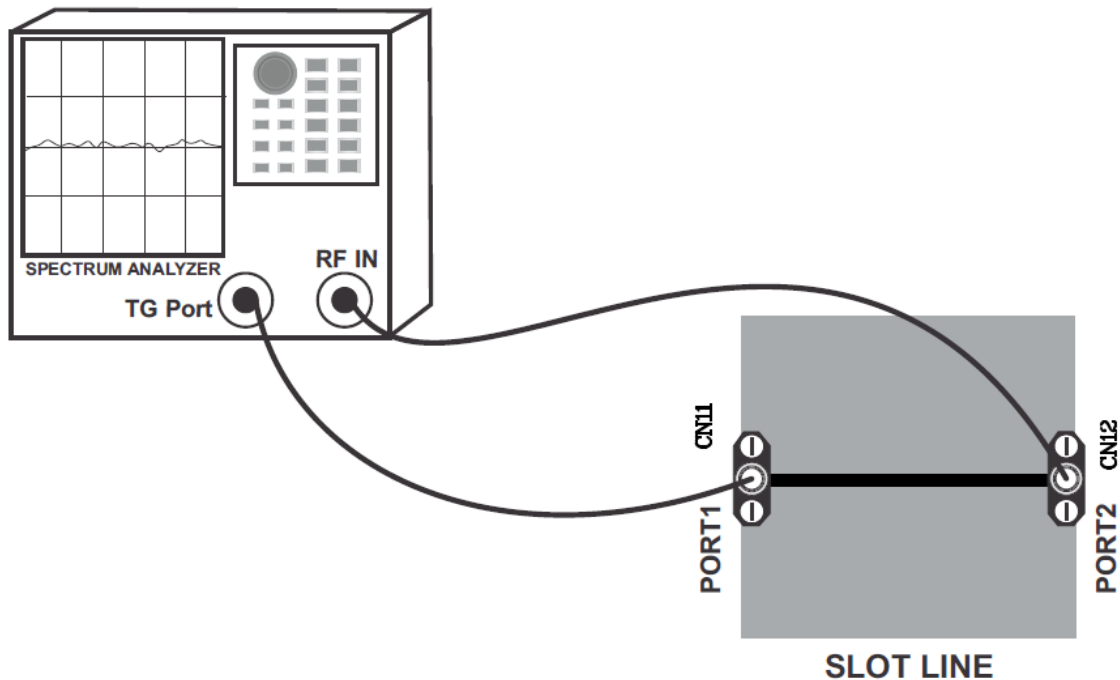
A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21 & S12) Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.



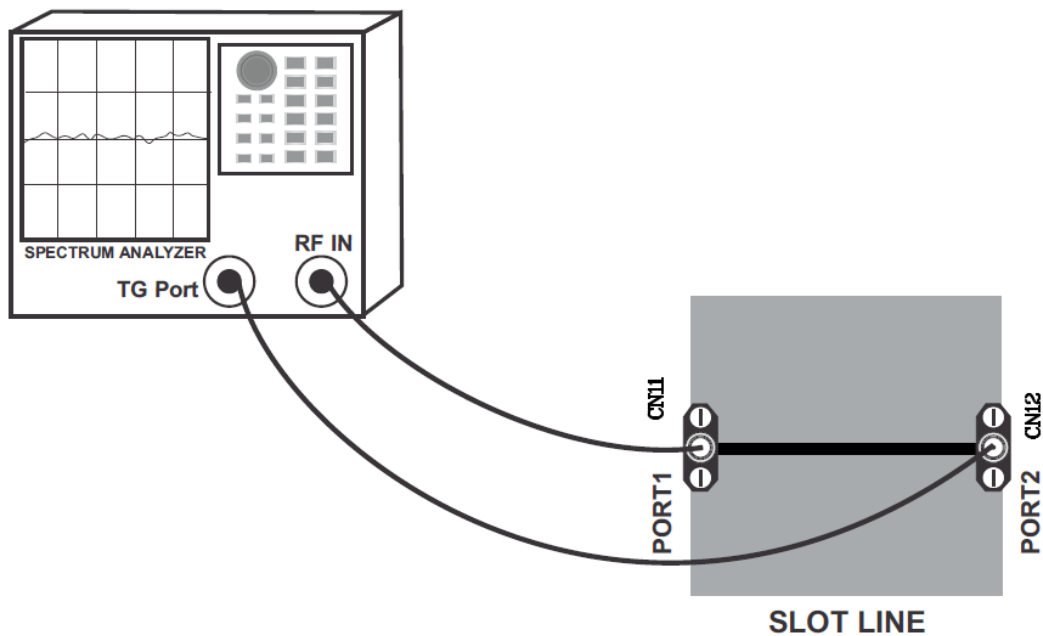
Setup for insertion loss (S_{21})

Measurement of Insertion Loss characteristics (S_{21}):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN11 of the Slot Line module (i.e. signal of the Spectrum analyzer is applied as input to the Slot Line)
- 2) Connect the CN12 of the Slot Line module to the Spectrum analyzer Port 2 (i.e. output from the Slot Line is applied to the VNA for data display).
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Measurement of Insertion loss characteristics (S_{12}):

- 1) Connect the output of the Spectrum analyzer (Port1) to the CN12 of the Slot Line module (i.e. signal of the Spectrum analyzer is applied as input to the Slot Line)
- 2) Connect the CN11 of the Slot Line module to the Spectrum analyzer Port 2 (i.e. output from the Slot Line is applied to the VNA for data display).



- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Save trace data for post processing and save trace image.

Reverence Plot (S21):



EXPERIMENT NO.13

EXPERIMENT NO.13

CHARACTERISATION OF OPEN AIR.

AIM:-

To understand the purpose and operational principle of the OPEN AIR circuits by using the RFL-RFT-C modules

OBJECTIVES:-

To explore the properties of Open Air,

EQUIPMENTS-

- RFL-RFT-C KIT
- Spectrum Analyzer
- 2 Monopole Helical Antenna.

SET UP & PROCEDURE:-

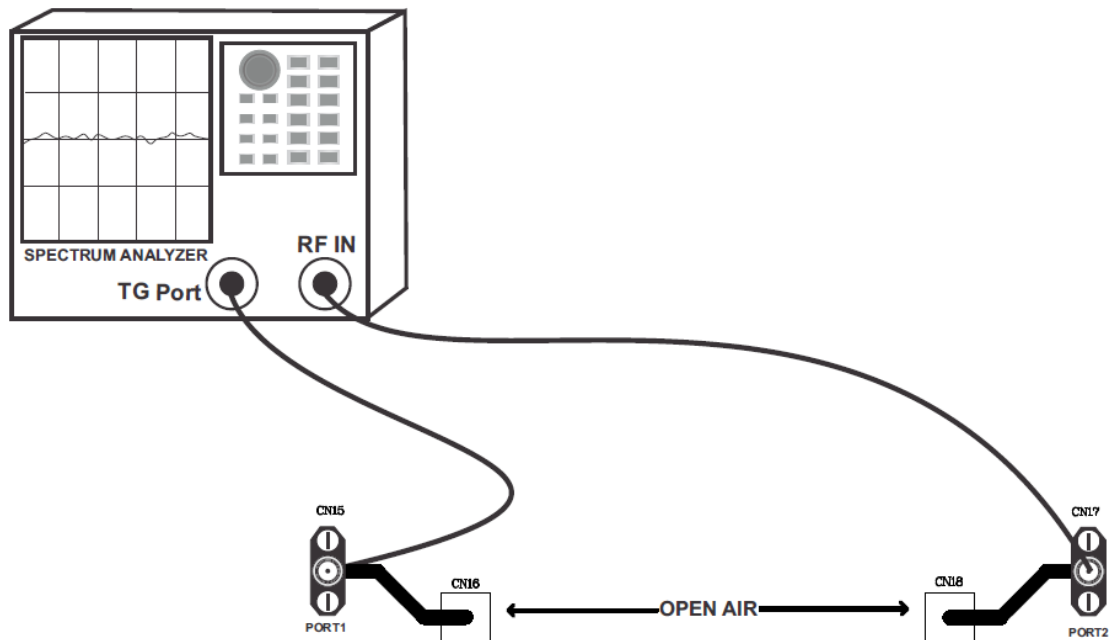
A. Using Spectrum Analyzer

Measurement of Co and Cross Polarization Characteristics

Procedure:-

- Frequency Range : 100 MHz to 1.3 GHz
- Power Level : 0 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, calibration should be performed again for the new settings or the data displayed may not be reliable.



Setup for Co-polarization (S21)

Measurement of Co-polarization characteristics (S21):

- 1) Connect Helical Antenna to CN 16 & CN 18.
- 2) Mount both Monopole Helical Antenna on Vertical position.
- 3) Connect the output of the Spectrum analyzer (Port1) to the CN15 of the Open Air module (i.e. signal of the Spectrum analyzer is applied as input to the Open Air)
- 4) Connect the CN17 of the Open Air module to the Spectrum Analyzer Port 2 (i.e. output from the Open Air is applied to the Spectrum analyzer for data display).
- 5) Apply markers and note down the readings for the frequencies specified in the observation table.
- 6) Save trace data for post processing and save trace image.

Measurement of Cross-polarization characteristics (S21):

- 1) Change the orientation of port 2 monopole antenna CN 18 on horizontal position.
- 2) Connect the output of the Spectrum analyzer (Port1) to the CN15 of the Open Air module (i.e. signal of the Spectrum analyzer is applied as input to the Open Air)
- 3) Connect the CN17 of the Open Air module to the Spectrum analyzer Port 2 (i.e. output from the Open Air is applied to the Spectrum analyzer for data display)
- 4) Apply markers and note down the readings for the frequencies specified in the observation table.
- 5) Save trace data for post processing and save trace image.

Reference Plot (Co-Polarization):



Reference Plot (Cross-Polarization):

